The Information Theory of AI Architectures

Effective Computational Capacity and Intelligence Decomposition

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Abstract

We define *Effective Computational Capacity* (ECC) as the generic Jacobian rank of the realization map from parameters to functions. This quantity measures the number of functionally independent parameter directions of a neural architecture, and we prove that it is the unique capacity measure satisfying three natural axioms: equivalence-invariance, direct-sum additivity, and normalization. Building on [1], we compute corrected ECC formulas for standard transformer components [13]. For full multi-head attention with H heads, head dimension d_h, and value dimension d_v,

$$\boxed{\mathrm{ECC}_{\mathrm{Attn}} = H\Big[d_h(2d - d_h) + d_v(2d - d_v)\Big]},$$

which specializes under the standard partition $d = Hd_h = Hd_v$ to

$$\boxed{\mathrm{ECC}_{\mathrm{Attn}} = 4d^2 - \frac{2d^2}{H}}.$$

For a width-m ReLU feed-forward network,

$$\boxed{\mathrm{ECC}_{\mathrm{FFN}} = m(2d - 1)},$$

so the standard choice $m = 4d$ yields $\mathrm{ECC}_{\mathrm{FFN}} = 8d^2 - 4d$. Thus, for individual transformer components, ECC is close to raw parameter count: most parameters are functionally active at first order. Compositional (multi-layer) ECC is upper-bounded by the sum of per-layer ECCs, and the main bottleneck is not static capacity but *utilization*.

To formalize this point we define task intelligence by

$$\boxed{\mathrm{Intelligence}(T) = \mathrm{ECC} \cdot \tau_{\mathrm{rep}}\tau_{\mathrm{stat}}\tau_{\mathrm{opt}}},$$

where τ_{rep} is representational alignment, τ_{stat} statistical efficiency, and τ_{opt} optimization quality. The operational identity

$$\boxed{\tau_{\mathrm{rep}} = \eta_{\mathrm{eff}} = \frac{k_{\mathrm{use}}}{\mathrm{ECC}}}$$

shows that $\mathrm{ECC}\,\tau_{\mathrm{rep}}$ is the usable task-aligned dimension. We then prove a conditional coding theorem linking excess risk to an aligned approximation dimension k, and we isolate the exact point at which a purely static bridge from ECC to performance fails. The correct local bridge is a measurable residual-tail quantity $\eta_T(k)$ derived from the pullback task metric $M_T = J_*^\top G_T J_*$. At a larger scale, performance obeys a six-term decomposition separating existence, geometry, slice selection, reachability, generalization, and optimization.

Applications include: a full ECC calculation for transformer attention including value and output maps, a function-space derivation of FFN capacity, a proof that standard Mamba blocks have $\mathrm{ECC} = \frac{1599}{256}d^2 + O(d) \approx 6.246\,d^2 + O(d)$, a geometric account of LoRA and catastrophic forgetting, a workload-aware compression and data-curation principle, a formal coverage-efficiency split for universal versus specialized models, and a no-universal-τ_{rep} theorem. In slogan form: [1] defined the ceiling; the present paper measures how much of that ceiling is actually used.

1 Introduction

Parameter count is the standard proxy for neural capacity, but it is a poor one. It does not distinguish active from inactive directions, it does not explain why different architectures with similar parameter counts behave differently, and it offers little guidance about why scaling laws, low-rank adaptation, sparse attention, and catastrophic forgetting often admit a common geometric structure. More fundamentally, it does not provide a mathematical definition of intelligence.

This paper proposes such a definition. The central object is the realization map

$$\Psi : \Theta \to \mathcal{F},$$

which sends a parameter vector to the function computed by the architecture on its input domain. We measure the functional size of an architecture not by $\dim \Theta$ but by the generic rank of $D\Psi$. This quantity, which we call *Effective Computational Capacity*, counts the number of functionally independent parameter directions. It removes gauge symmetries, dead directions, and other forms of overparameterized redundancy by construction.

The paper builds directly on [1], which identified the d^2 score-side ceiling of bilinear attention and showed that degree-2 attention is Pareto-optimal on the score side. The present work extends that program in two directions.

First, we turn ECC into a general architectural invariant. We prove that $\mathrm{ECC}(\Psi) = \mathrm{rank}_{\mathrm{gen}}(D\Psi)$ is the unique capacity measure satisfying three natural axioms, and we compute ECC for full multi-head attention, ReLU feed-forward layers, and Mamba-style selective state-space models.

Second, we argue that capacity is only one factor in performance. For modern architectures, individual component ECCs (attention, FFN) are already close to raw parameter count, and compositional ECC is at most their sum. The central question is therefore not "How much capacity exists?" but "How much of that capacity is aligned with the task, statistically identified, and actually reached by training?" This leads to the master equation

$$\mathrm{Intelligence}(T) = \mathrm{ECC} \cdot \tau_{\mathrm{rep}} \tau_{\mathrm{stat}} \tau_{\mathrm{opt}}.$$

The multiplicative factor $\tau_{\mathrm{rep}} \tau_{\mathrm{stat}} \tau_{\mathrm{opt}} < 1$ is the real bottleneck. The waste is not primarily in raw capacity but in alignment.

The paper is organized around that distinction. Sections 2 and 3 define ECC and prove its uniqueness. Section 4 computes corrected ECC formulas for transformer components and discusses the still-open multi-layer additivity problem. Section 5 introduces the intelligence decomposition and formalizes usable task-aligned dimension. Section 6 gives a conditional coding theorem, shows why a purely static bridge from ECC to performance is impossible in general, and then provides the correct local bridge via a measurable residual-tail quantity. Section 7 develops applications, including LoRA, forgetting, self-training, compression, and architecture screening. Section 8 studies future architectural directions, including complex-valued attention and the impossibility of a universally optimal representational bias. Section 9 presents numerical illustrations. Section 10 situates the framework relative to related work and major AI debates. Section 11 concludes.

Main contributions.

1. We prove that $\mathrm{ECC}(\Psi) = \mathrm{rank}_{\mathrm{gen}}(D\Psi)$ is the unique capacity measure satisfying equivalence-invariance, additivity, and normalization.
2. We derive corrected formulas for full multi-head attention and ReLU feed-forward layers:
$$\mathrm{ECC}_{\mathrm{Attn}} = H\big[d_h(2d - d_h) + d_v(2d - d_v)\big], \qquad \mathrm{ECC}_{\mathrm{FFN}} = m(2d - 1).$$
3. We resolve the multi-layer additivity question: the exact formula is $\mathrm{ECC}_{\mathrm{total}} = \dim(\mathcal{S}_1 + \cdots + \mathcal{S}_L)$, where $\mathcal{S}_l$ is the transported tangent image of layer l. Exact additivity ($= \sum \mathrm{ECC}_l$) holds if and only if the transported tangent spaces form a direct sum. We provide three counterexamples (smooth linear, ReLU residual, and attention at $T = 1$) and a Signature Criterion as a verifiable sufficient condition.

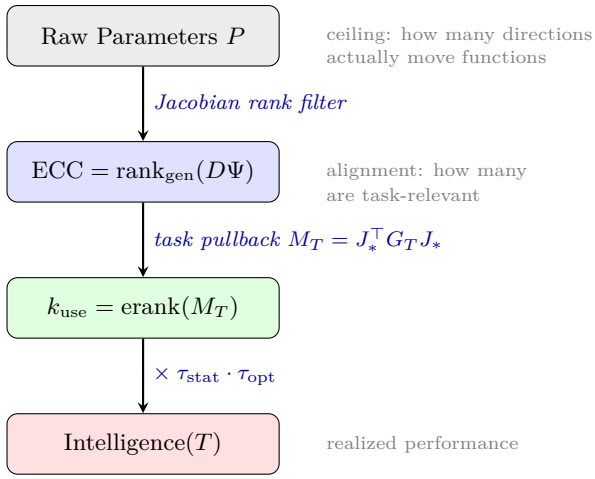

Figure 1: The information-theoretic pipeline: raw parameter count is filtered by Jacobian rank (ECC), then by task alignment (τ_{rep}), statistical identification (τ_{stat}), and optimization quality (τ_{opt}), yielding realized task intelligence.

4. We prove a conditional coding theorem and isolate a measurable local bridge to performance via the pullback task metric and its residual tail $\eta_T(k)$.
5. We define task intelligence as capacity times utilization, with an operational representational factor $\tau_{\mathrm{rep}} = k_{\mathrm{use}}/\,\mathrm{ECC}$.
6. We derive applications to LoRA, forgetting, self-training, architecture screening, workload-aware compression, data curation, emergence, and the technical tension between "scaling works" and "scaling wastes structure".
7. We prove that no single architecture can maximize expected representational efficiency across all domain mixtures once mismatch crossovers exist.

2 Preliminaries

2.1 Notation

Throughout, d denotes the model width, H the number of attention heads, $d_h = d/H$ the head dimension, d_v the value dimension, m the hidden width of a feed-forward network, n sequence length, and L the number of layers. We write $\Theta \subseteq \mathbb{R}^P$ for parameter space, $\mathcal{F}$ for a function space over a compact input domain $\mathcal{X}$, and $\Psi : \Theta \to \mathcal{F}$ for the realization map.

For transformers we use the standard forms

$$S(X; M) = \mathrm{softmax}\Big(\frac{XMX^\top}{\sqrt{d_h}}\Big), \qquad \mathrm{Attn}(X) = S(X; M)\, XV,$$

for a single head, where M denotes the score kernel and V a value map. Multi-head attention concatenates the head outputs and multiplies by an output map W_O. A standard feed-forward block has the form

$$\mathrm{FFN}(x) = W_2\sigma(W_1 x + b_1) + b_2,$$

with $\sigma = \mathrm{ReLU}$ unless stated otherwise.

Standing manifold convention. Unless explicitly stated otherwise, the realized-function space $\mathcal{F}$ is viewed as a smooth Banach (or Fréchet) manifold, typically an open subset of $C^r(\mathcal{X}, \mathbb{R}^q)$ or a smooth finite-dimensional submanifold thereof. Every realization map $\Psi : \Theta \to \mathcal{F}$ considered in Sections 2–6 is smooth in this manifold sense. All statements about local equivalence of realized families refer to equivalence of germs inside this ambient smooth manifold.

2.2 Background facts

We use the following standard tools.

- **Jacobian rank.** For a smooth map $f : \mathbb{R}^m \to \mathbb{R}^n$, the rank of Df_x is the dimension of the image of the tangent map at x.
- **Constant-rank theorem.** If $f : M \to N$ is smooth between manifolds and $\operatorname{rank} Df$ is constant on an open set $U \subset M$, then for every $x \in U$ there are neighborhoods $x \in U_x \subset U$ and $f(x) \in V_x \subset N$ such that $f(U_x) \cap V_x$ is a smoothly embedded submanifold of dimension $\operatorname{rank} Df|_U$. In particular, the image is *locally* a smooth submanifold of that dimension.
- **Manifold width and nonlinear approximation.** We write $\delta_k(K)$ for a k-parameter approximation width of a compact family K. The exact choice is not central; the relevant point is that approximation error and intrinsic dimensionality are different notions.
- **Generalization complexity.** Standard entropy or Rademacher arguments imply that a k-parameter Lipschitz family behaves, from the viewpoint of finite-sample complexity, like a k-dimensional model up to logarithmic factors.

2.3 Relation to the d^2 pullback theorem

Reference [1] showed that bilinear attention has a d^2-dimensional score-side ceiling when the score kernel is treated as a free bilinear operator. That paper therefore identified the natural capacity scale of attention itself. The present paper lifts that ceiling through concrete architectural parameterizations. In particular, once the score kernel is factored through W_Q and W_K, the realized family becomes a low-rank matrix variety, and the relevant dimension becomes the rank-variety dimension $d_h(2d - d_h)$. Thus [1] supplies the score-side ceiling, while the present paper computes the realized architectural capacity after parameterization, extension to values and outputs, and comparison with other modules.

3 Effective Computational Capacity

3.1 Definition

Definition 3.1 (Pointwise ECC)**.** *Let $\Psi : \Theta \to \mathcal{F}$ be smooth. The effective computational capacity at $\theta \in \Theta$ is*

$$\mathrm{ECC}(\Psi, \theta) := \operatorname{rank}(D\Psi_\theta).$$

Definition 3.2 (Architecture-level ECC)**.** *The effective computational capacity of the architecture is the generic rank*

$$\boxed{\mathrm{ECC}(\Psi) := \operatorname{rank}_{\mathrm{gen}}(D\Psi) = \max_{\theta \in \Theta} \operatorname{rank}(D\Psi_\theta).}$$

For real-analytic realization maps, the maximum is attained on an open dense subset of Θ.

The pointwise quantity measures the number of first-order parameter directions that genuinely move the realized function. The generic quantity removes nongeneric singularities and serves as an architectural invariant.

3.2 Axiomatic uniqueness

The key question is not whether Jacobian rank is *a* reasonable definition, but whether it is *the* unavoidable one once a few natural requirements are imposed.

Theorem 3.3 (Uniqueness of ECC)**.** *Let C assign a nonnegative real number to every smooth realization map $\Psi : \Theta \to \mathcal{F}$. Assume:*

(A1) ***Equivalence-invariance.*** *If $\Psi_1 : \Theta_1 \to \mathcal{F}_1$ and $\Psi_2 : \Theta_2 \to \mathcal{F}_2$ have locally smoothly equivalent realized-family germs at generic regular points—that is, there exist regular points $\theta_i \in \Theta_i$, neighborhoods $U_i \ni \theta_i$, neighborhoods $V_i \ni \Psi_i(\theta_i)$, and a diffeomorphism $\chi : V_1 \to V_2$ such that*

$$\chi\big(\Psi_1(U_1)\big) = \Psi_2(U_2),$$

—then $C(\Psi_1) = C(\Psi_2)$.

(A2) ***Additivity under external direct sums.*** *If*

$$\Psi = \Psi_1 \boxplus \Psi_2 : \Theta_1 \times \Theta_2 \to \mathcal{F}_1 \times \mathcal{F}_2, \qquad \Psi(\theta_1, \theta_2) = (\Psi_1(\theta_1), \Psi_2(\theta_2)),$$

with disjoint parameter spaces and split codomain, then

$$C(\Psi) = C(\Psi_1) + C(\Psi_2).$$

(A3) ***Normalization.*** *For the identity map $\mathrm{id} : \mathbb{R} \to \mathbb{R}$, one has $C(\mathrm{id}) = 1$.*

Then

$$C(\Psi) = \mathrm{rank}_{\mathrm{gen}}(D\Psi) = \mathrm{ECC}(\Psi)$$

for every smooth realization map.

Proof. Let

$$r = \mathrm{rank}_{\mathrm{gen}}(D\Psi).$$

By the constant-rank theorem, near any generic regular point of rank r the realized-family germ of Ψ is locally smoothly equivalent to the germ of the coordinate projection

$$\pi_r : \mathbb{R}^p \to \mathbb{R}^r, \qquad (x_1, \ldots, x_p) \mapsto (x_1, \ldots, x_r),$$

and hence to the germ of $\mathrm{id}_{\mathbb{R}^r}$ (with the convention $\mathbb{R}^0 = \{0\}$ when $r = 0$). By (A1), $C(\Psi)$ therefore depends only on r. Thus there exists a function

$$\varphi : \mathbb{N}_0 \to \mathbb{R}_{\geq 0}$$

such that

$$C(\Psi) = \varphi(r).$$

Now apply (A2) to

$$\mathrm{id}_{\mathbb{R}^{r_1}} \boxplus \mathrm{id}_{\mathbb{R}^{r_2}} \simeq \mathrm{id}_{\mathbb{R}^{r_1+r_2}}\,.$$

We obtain

$$\varphi(r_1 + r_2) = \varphi(r_1) + \varphi(r_2), \qquad (r_1, r_2 \in \mathbb{N}_0).$$

Setting $r_1 = r_2 = 0$ gives

$$\varphi(0) = \varphi(0) + \varphi(0),$$

hence

$$\varphi(0) = 0.$$

By (A3),

$$\varphi(1) = 1.$$

An induction on r then yields

$$\varphi(r) = r \qquad \text{for all } r \in \mathbb{N}_0.$$

Therefore

$$C(\Psi) = \varphi(r) = r = \operatorname{rank}_{\mathrm{gen}}(D\Psi) = \mathrm{ECC}(\Psi).$$

□

Remark 3.4 (Naturalness of the axioms). The axioms are intentionally minimal. Equivalence-invariance says that two architectures realizing the same local function family should have the same capacity. Additivity says that independent subsystems should contribute additively. Normalization fixes the unit. None of these axioms mentions Jacobian rank, so rank appears as a theorem rather than an assumption. This is structurally parallel to classical uniqueness theorems for entropy.

3.3 Parameter count versus functional count

Let $P = \dim \Theta$ denote raw parameter count. Then always

$$\mathrm{ECC}(\Psi) \leq P.$$

The difference

$$P - \mathrm{ECC}(\Psi)$$

is the dimension of redundant directions: gauge symmetries, exact parameter non-identifiabilities, dead features, and other first-order null modes. Parameter count is therefore storage size. ECC is functional freedom.

This distinction is especially sharp in attention and feed-forward layers. Yet, as Section 4.6 shows, individual transformer components are unusual in that their per-component ECC is very close to parameter count; compositional ECC is upper-bounded by the sum. This is why utilization, not raw block capacity, becomes the dominant issue.

3.4 Relation to prior notions

Several earlier notions are close to ECC but not identical in role.

- **Functional dimension.** Grigsby, Lindsey, and Rolnick study the functional dimension of ReLU networks and relate it to batch NTK rank [3]. This is very close in spirit to ECC.
- **Neuromanifold dimension.** Henry, Marchetti, and Kohn compute the dimension of function spaces for lightning attention and related models [2]. Their quantity coincides with generic Jacobian rank in the cases they analyze.
- **Intrinsic dimension.** The intrinsic-dimension literature measures the dimensionality of optimization landscapes or low-dimensional training subspaces [11, 10]. This is a different object: it concerns trainability or optimization compression, not the realized function-family dimension.
- **Classical complexity notions.** VC dimension, Fisher rank, and NTK rank measure other aspects of learning. ECC should be viewed as complementary rather than substitutive.

3.5 Position relative to existing capacity notions

Several related capacity notions exist in the literature. *Intrinsic dimension* [11, 10] measures the effective dimensionality of a particular optimization trajectory, not the architectural capacity of the function space itself. *NTK rank* captures the linear approximation capacity in a neighborhood of initialization, but does

not account for the full nonlinear function space that training can explore. *Fisher rank* is conditioned on a specific data distribution and therefore is not an architecture invariant.

ECC is complementary to, not a replacement for, these notions. However, it is the *unique* capacity measure satisfying the three axioms of §3.2: equivalence-invariance, direct-sum additivity, and normalization (Theorem 3.3). This axiomatic characterization is what separates ECC from ad hoc capacity proxies and grounds the subsequent intelligence decomposition.

4 ECC of Transformer Blocks and Related Architectures

Remark 4.1 (Function-space category). For attention and gating mechanisms involving softmax or sigmoid, the realization map Ψ is real-analytic, and generic rank is attained on an open dense subset of Θ (Zariski-open in the algebraic-geometric sense). For ReLU networks, Ψ is piecewise linear and semialgebraic; the generic rank is attained on a Euclidean-open dense semialgebraic subset. In both cases the Jacobian $D\Psi$ and its rank are well-defined almost everywhere, and the generic-rank notion used throughout this section is the standard one for the respective category.

4.1 Single-head score-side capacity

We begin with the score kernel induced by a single attention head.

Theorem 4.2 (Single-head score-side ECC)**.** *Let*

$$\Phi(W_Q, W_K) = W_Q W_K^\top, \qquad W_Q, W_K \in \mathbb{R}^{d\times d_h}.$$

Then

$$\boxed{\mathrm{ECC}_{\mathrm{score}} = d_h(2d - d_h).}$$

In the square case $d_h = d$, this reduces to $\mathrm{ECC}_{\mathrm{score}} = d^2$.

Proof. The map Φ parametrizes the variety of $d \times d$ matrices of rank at most d_h. For generic full-column-rank (W_Q, W_K), the fiber is exactly the right action of $\mathrm{GL}(d_h)$:

$$(W_Q, W_K) \mapsto (W_Q G,\ W_K G^{-\top}).$$

Hence the generic fiber dimension is d_h^2. Since the domain has dimension $2dd_h$, the image dimension is

$$2dd_h - d_h^2 = d_h(2d - d_h).$$

□

Remark 4.3. The theorem lifts the score-side ceiling of [1] through the concrete $QK^\top$ factorization. The difference between d_h^2 and $d_h(2d - d_h)$ is the difference between treating the bilinear kernel as a free operator and treating it as a low-rank factorized architectural family.

4.2 Full multi-head attention

The key correction of this paper is that full attention capacity is not exhausted by the score kernel alone. The value and output maps contribute their own functional dimensions.

Theorem 4.4 (Full MHA ECC)**.** *Consider multi-head attention with H heads, query-key factors $W_Q^h, W_K^h \in \mathbb{R}^{d\times d_h}$, value factors $W_V^h \in \mathbb{R}^{d\times d_v}$, and output blocks $O_h \in \mathbb{R}^{d_v\times d}$, evaluated on input sequences of length $n \geq 2$. Then the generic ECC of the attention module is*

$$\boxed{\mathrm{ECC}_{\mathrm{Attn}} = H\big[d_h(2d - d_h) + d_v(2d - d_v)\big].}$$

Under the standard partition $d = Hd_h = Hd_v$,

$$\boxed{\mathrm{ECC}_{\mathrm{Attn}} = 4d^2 - \frac{2d^2}{H}.}$$

With query, key, value, and output biases,

$$\boxed{\mathrm{ECC}_{\mathrm{Attn+bias}} = H\big[d_h(2d - d_h + 1) + d_v(2d - d_v)\big] + d.}$$

Proof. For each head, define

$$M_h = W_Q^h (W_K^h)^\top, \qquad N_h = W_V^h O_h.$$

The score-side factorization contributes $d_h(2d - d_h)$ by Theorem 4.2. The value-output factorization is another rank-d_v matrix variety and contributes $d_v(2d - d_v)$ by the same argument.

What remains is to show that these tangent directions are generically independent after passing through softmax, concatenation, and output composition. On a generic two-token slice of the input space, score perturbations change the nonlinear attention kernels, whereas value-output perturbations change the transported values through different analytic coefficient functions. Distinct heads produce distinct nonlinear kernels, and the reduced differential is generically immersive. Therefore the contributions add across the score side, the value-output side, and across heads. The full argument, including an explicit confluent Vandermonde injectivity proof, is given in Lemma A.1 (Appendix A). □

Remark 4.5 (Bias terms). The bias contributions are asymmetric. Query bias contributes $+d_h$ per head, because it genuinely perturbs the score matrix. Key bias is generically invisible because row-wise softmax is invariant under adding the same constant to every key score in a row. Value bias contributes only through a constant-output block, and output bias contributes a global d-dimensional constant block.

4.3 ReLU feed-forward networks

The second major correction is that the ECC of a ReLU feed-forward layer is not the pointwise Jacobian rank inside a fixed activation region. Capacity must be computed in function space, and activation-boundary motion contributes additional degrees of freedom.

Theorem 4.6 (FFN ECC)**.** *Consider a one-hidden-layer ReLU network*

$$f(x) = W_2\sigma(W_1 x), \qquad W_1 \in \mathbb{R}^{m\times d}, \quad W_2 \in \mathbb{R}^{d\times m},$$

with $d \geq 2$. Then

$$\boxed{\mathrm{ECC}_{\mathrm{FFN}} = m(2d - 1).}$$

For the standard transformer choice $m = 4d$,

$$\boxed{\mathrm{ECC}_{\mathrm{FFN}} = 8d^2 - 4d.}$$

If hidden and output biases are included, then

$$\mathrm{ECC}_{\mathrm{FFN+bias}} = 2md + d.$$

Proof. There are $2md$ parameters in (W_1, W_2). Each hidden neuron has a one-dimensional positive-scaling gauge:

$$(c_i, a_i) \mapsto (\lambda c_i, \lambda^{-1} a_i), \qquad \lambda > 0,$$

which leaves the realized function unchanged because ReLU is positively homogeneous. Hence $\dim\ker D\Psi \geq m$.

To show that this is the full kernel, one must work in function space rather than at a single input point. Inside one activation cone, the pointwise rank can be at most d, which is far too small. The missing directions

appear as movement of the activation hyperplanes. For generic parameters and $d \geq 2$, the distributional jumps across the neuron hyperplanes $H_i = \{x : a_i^\top x = 0\}$ isolate each neuron separately. A tangent vector in the kernel must therefore preserve both the output amplitudes and the hyperplane motion of every neuron, forcing it to lie in the direct sum of the m homogeneity gauges. Hence the generic kernel has dimension exactly m, and

$$\mathrm{ECC}_{\mathrm{FFN}} = 2md - m = m(2d - 1).$$

Appendix B contains the detailed argument. □

Remark 4.7. The often-quoted expression $\min(m, d^2)$ is a pointwise rank bound inside a fixed activation region. It is not the function-space ECC. The difference is precisely the freedom to move activation boundaries.

4.4 Multi-layer composition and the open additivity problem

Let an L-layer residual network be written schematically as

$$h_l = h_{l-1} + g_l(h_{l-1}; \theta_l).$$

For each layer, transport its tangent family to function space at the network output and denote the resulting subspace by $\mathcal{S}_l(\theta)$.

Proposition 4.8 (General composition formula)**.** *At any regular parameter point θ,*

$$\boxed{\mathrm{ECC}_{\mathrm{total}}(\theta) = \dim\big(\mathcal{S}_1(\theta) + \cdots + \mathcal{S}_L(\theta)\big).}$$

This is the correct formula. It immediately shows that exact additivity is a question about the geometry of the transported tangent subspaces, not just about the sum of local layer ranks.

The current state of the problem is genuinely mixed, so both sides must be stated.

Proposition 4.9 (Conditional additivity under singular-support separation)**.** *Suppose each nonzero function in $\mathcal{S}_l(\theta)$ carries a singular part supported on a set Σ_l, and suppose:*

(i) all prefix and suffix transports are generically invertible, so $\dim \mathcal{S}_l(\theta) = \mathrm{ECC}_l$;

(ii) the singular-support sets $\Sigma_1, \ldots, \Sigma_L$ are pairwise disjoint;

(iii) no nonzero element of $\mathcal{S}_l(\theta)$ becomes smooth near every point of Σ_l.

Then

$$\mathcal{S}_1(\theta) \oplus \cdots \oplus \mathcal{S}_L(\theta)$$

is a direct sum, and therefore

$$\mathrm{ECC}_{\mathrm{total}}(\theta) = \sum_{l=1}^{L} \mathrm{ECC}_l\,.$$

Proof. Assume $\sum_{l=1}^{L} f_l = 0$ with $f_l \in \mathcal{S}_l(\theta)$. Fix j and localize near a point of Σ_j that lies outside all other Σ_ℓ. By construction, all f_ℓ for $\ell \neq j$ are smooth there, whereas any nonzero f_j retains a singular component. Since the sum vanishes identically, its singular part must vanish, so f_j is smooth near every point of Σ_j. By assumption (iii), this implies $f_j = 0$. Repeating over j yields $f_l = 0$ for all l. □

The preceding proposition captures the intuition behind why residual piecewise-linear networks may behave additively when different layers create distinguishable singular structures.

However, additivity fails in general.

Proposition 4.10 (Scalar residual counterexample)**.** *On the Euclidean-open parameter set*

$$U := \{(a_1, \dots, a_L) \in \mathbb{R}^L : a_l \neq -1 \text{ for all } l\},$$

consider the scalar residual layers

$$f_l(h; a_l) = (1 + a_l)h,$$

Each layer has pointwise rank 1 *on* U*, but the* L*-layer composition*

$$F(x) = \prod_{l=1}^{L} (1 + a_l)x$$

has

$$\mathrm{ECC}_{\mathrm{total}} = 1 \neq L.$$

Proof. For a single layer,

$$\partial_{a_l} f_l(h; a_l) = h,$$

so its pointwise Jacobian rank is 1 whenever $a_l \neq -1$. For the composition,

$$\frac{\partial F}{\partial a_l}(x) = \Big(\prod_{j \neq l} (1 + a_j)\Big)x.$$

On U, each derivative is a nonzero scalar multiple of the same function x. Hence all Jacobian columns are collinear, so the Jacobian rank is 1. □

Proposition 4.11 (ReLU residual counterexample)**.** *Consider the family on the Euclidean-open parameter set*

$$U := \{(a_1, \dots, a_L) \in \mathbb{R}^L : a_l > -1 \text{ for all } l\}$$

given by

$$r_{a_l}(x) = x + a_l\, e_1(e_1^\top x)_+, \qquad l = 1, \dots, L.$$

Each layer has $\mathrm{ECC}_l = 1$ *on* U*. However, on* U *the composed network has*

$$\mathrm{ECC}_{\mathrm{total}} = 1 \neq L.$$

Proof. For a single layer,

$$\partial_{a_l} r_{a_l}(x) = e_1(e_1^\top x)_+ \neq 0,$$

so $\mathrm{ECC}_l = 1$. If $a_l > -1$, then the half-space $\{e_1^\top x > 0\}$ is forward-invariant:

$$e_1^\top r_{a_l}(x) = (1 + a_l)e_1^\top x > 0 \qquad \text{whenever } e_1^\top x > 0,$$

while points with $e_1^\top x \leq 0$ are fixed. Hence, for x with $e_1^\top x > 0$,

$$R(x) := r_{a_L} \circ \cdots \circ r_{a_1}(x) = x + \Big(\prod_{l=1}^{L} (1 + a_l) - 1\Big)e_1(e_1^\top x),$$

and for $e_1^\top x \leq 0$ one has $R(x) = x$. Therefore

$$\frac{\partial R}{\partial a_l}(x) = \Big(\prod_{j \neq l} (1 + a_j)\Big)e_1(e_1^\top x)_+,$$

so all Jacobian columns are proportional to the same base function $e_1(e_1^\top x)_+$. Thus the Jacobian rank of the composed family is 1 on U. □

Proposition 4.12 (Attention counterexample at $T = 1$)**.** *For sequence length $T = 1$, a multi-layer attention residual network reduces to a smooth linear residual network because* softmax *on a 1×1 matrix yields the constant 1, and all score-side parameters become invisible. The problem then reduces to a linear residual composition, and in general $\mathrm{ECC}_{\mathrm{total}} < L \cdot \mathrm{ECC}_1$: the transported tangent spaces overlap nontrivially even though individual layers have full single-layer ECC. For example, when $d = 1$ this reduces to the scalar residual case of Proposition 4.10, for which $\mathrm{ECC}_{\mathrm{total}} = 1 < L$.*

These three counterexamples (scalar linear, ReLU residual, and attention at $T = 1$) show that additivity failure is not a pathology of artificially simple models but occurs across smooth, piecewise-linear, and nonlinear architectures.

Example 4.13 (Attention–FFN additivity is context-dependent)*.* Even when block *types* differ, additivity depends on the surrounding layers, not on block type alone.

Case 1 (additivity holds). Take $T = d = 1$, a residual attention block $a_\alpha(h) = (1 + \alpha)h$ with $\alpha > -1$, and a residual ReLU block $r_b(h) = h + b\,h_+$. Then $F = r_b \circ a_\alpha$ gives

$$F(x) = (1 + \alpha)x_- + (1 + \alpha)(1 + b)x_+.$$

The derivatives

$$\partial_\alpha F = x_- + (1 + b)x_+, \qquad \partial_b F = (1 + \alpha)x_+$$

are linearly independent as functions. Hence $\mathrm{ECC}_{\mathrm{total}} = 2 = \mathrm{ECC}_1 + \mathrm{ECC}_2$.

Case 2 (additivity fails). Insert a preceding ReLU truncation $p(x) = x_+$. Now $F = r_b \circ a_\alpha \circ p$ gives

$$F(x) = (1 + \alpha)(1 + b)\, x_+,$$

so $\partial_\alpha F = (1 + b)\, x_+$ and $\partial_b F = (1 + \alpha)\, x_+$ are proportional. Hence $\mathrm{ECC}_{\mathrm{total}} = 1 < 2$.

Conclusion: Block-type diversity (attention versus FFN) does *not* guarantee additivity. The full compositional context determines whether the transported tangent spaces are independent.

Proposition 4.14 (Signature Criterion for additivity)**.** *Suppose there exist linear operators $T_l : \mathcal{F} \to \mathcal{F}$ such that:*

(i) $T_l|_{\mathcal{S}_l(\theta)}$ is injective;

(ii) $T_l(\mathcal{S}_j(\theta)) = \{0\}$ for all $j \neq l$.

Then $\mathcal{S}_1(\theta) \oplus \cdots \oplus \mathcal{S}_L(\theta)$ is a direct sum, and $\mathrm{ECC}_{\mathrm{total}}(\theta) = \sum_{l=1}^{L} \mathrm{ECC}_l$.

Proof. If $\sum_l f_l = 0$ with $f_l \in \mathcal{S}_l(\theta)$, applying T_j yields $T_j(f_j) = 0$. Injectivity of T_j on $\mathcal{S}_j$ then forces $f_j = 0$. □

Remark 4.15 (Resolution of the additivity question)*.* The general composition formula $\mathrm{ECC}_{\mathrm{total}} = \dim(\mathcal{S}_1 + \cdots + \mathcal{S}_L)$ is the correct and always-true statement. Exact additivity ($= \sum \mathrm{ECC}_l$) is a special case requiring transversality of the transported tangent spaces, which can be certified by the Signature Criterion above. The singular-support separation of Proposition 4.9 is one instance, applicable to piecewise-linear residual networks with well-separated activation boundaries, but it does not extend to smooth blocks such as softmax attention or GELU activations. For practical multi-layer transformer blocks, the compositional ECC should be understood as $\dim(\mathcal{S}_1 + \cdots + \mathcal{S}_L)$, with the per-block sum $\sum \mathrm{ECC}_l$ serving as an upper bound.

4.5 Mamba and selective state-space models

Theorem 4.16 (Mamba ECC)**.** *Consider a standard Mamba-style selective state-space block [14] in the concrete Mamba-v1 parameterization formalized in Appendix C, with model width d, inner width m, state*

size N, step-size rank r, and convolution width k. For sufficiently long input sequences ($L \geq \max(2, k)$), the generic ECC of the full block is

$$\mathrm{ECC}_{\mathrm{Mamba}} = 3dm + m(k + 3N + 2r + 3) - r^2 - N.$$

Under the usual defaults $m = 2d$, $N = 16$, $k = 4$, and $r = \lceil d/16 \rceil$,

$$\boxed{\mathrm{ECC}_{\mathrm{Mamba}} = \tfrac{1599}{256}\, d^2 + O(d) \approx 6.246\, d^2 + O(d).}$$

Proof. The parameter count of the official block is

$$P = 3dm + m(k + 3N + 2r + 3).$$

Two generic symmetry classes reduce the rank. First, the low-rank factorization used for the step-size channel has a $\mathrm{GL}(r)$ gauge, contributing an r^2-dimensional kernel. Second, the latent state coordinates admit an N-dimensional rescaling symmetry between state input and state output factors. For generic sequence length $L \geq \max(2, k)$ these are the only persistent first-order null modes, yielding

$$\mathrm{ECC} = P - (r^2 + N).$$

The dependence on L saturates once the convolution and state-transition parameters become visible. Appendix C spells out the Mamba-v1 parameterization, proves the two non-structural hypotheses used in the kernel calculation on a Euclidean-open dense subset of parameter space, and then identifies the full kernel exactly. □

4.6 Architectural comparison

Table 1 compares the main building blocks.

Architecture	ECC (no bias)	Standard width formula	Compute scale
Attention (H heads)	$H[d_h(2d - d_h) + d_v(2d - d_v)]$	$4d^2 - 2d^2/H$	$O(n^2 d)$
FFN (ReLU, width m)	$m(2d - 1)$	$8d^2 - 4d$ for $m = 4d$	$O(nd^2)$
Transformer block	$\leq \mathrm{ECC}_{\mathrm{Attn}} + \mathrm{ECC}_{\mathrm{FFN}}$	$\leq 12d^2 - 2d^2/H - 4d$	$O(n^2 d + nd^2)$
Mamba block	$3dm + m(k + 3N + 2r + 3) - r^2 - N$	$\frac{1599}{256} d^2 + O(d)$	$O(nd^2)$

Table 1: Corrected ECC scales for the main architectural blocks. The transformer block line is an upper bound; exact additivity holds when the Signature Criterion (Proposition 4.14) is satisfied.

For the standard case $H = 8$, the block upper bound becomes

$$\mathrm{ECC}_{\mathrm{block}} \leq 11.75d^2 - 4d.$$

Since a standard transformer block has approximately $12d^2$ parameters excluding biases, the sum of per-component ECCs is numerically close to parameter count and serves as an upper bound on compositional block ECC. This is the main empirical message of the corrected formulas: at component scale, modern transformer parameters are not mostly dead. The real inefficiency lies in task alignment, not static first-order capacity.

5 Intelligence as Capacity Times Utilization

5.1 Master equation

We now formalize the central claim of the paper.

Definition 5.1 (Task intelligence)**.** *For an architecture-task pair, define*

$$\boxed{\text{Intelligence}(T) = \text{ECC} \cdot \bar{\tau}_\Sigma, \qquad \bar{\tau}_\Sigma := \tau_{\text{rep}} \tau_{\text{stat}} \tau_{\text{opt}}.}$$

This is not a metaphysical definition of all aspects of intelligence. It is a functional and task-relative definition: how much of the architecture's ceiling is aligned to the task, statistically identified, and actually realized by optimization.

5.2 Exact telescoping decomposition

Proposition 5.2 (Utilization factorization)**.** *Let*

- k_{use} *be the usable task-aligned dimension,*
- k_{obs} *the portion identifiable from finite data,*
- k_{reached} *the portion actually reached by the optimizer.*

Define

$$\tau_{\text{rep}} := \frac{k_{\text{use}}}{\text{ECC}}, \qquad \tau_{\text{stat}} := \frac{k_{\text{obs}}}{k_{\text{use}}}, \qquad \tau_{\text{opt}} := \frac{k_{\text{reached}}}{k_{\text{obs}}}.$$

Then

$$\bar{\tau}_\Sigma = \tau_{\text{rep}} \tau_{\text{stat}} \tau_{\text{opt}} = \frac{k_{\text{reached}}}{\text{ECC}}.$$

Proof. Immediate by telescoping cancellation; the identity is definitional rather than explanatory, and the content lies in the operationalization of the intermediate dimensions. □

Remark 5.3. The decomposition is exact once the intermediate dimensions are fixed. The only modeling choice lies in how one operationalizes those dimensions.

5.3 The canonical local object: the pullback task metric

The pullback task metric

$$M_T(\theta^\star) = J_\star^\top G_T J_\star$$

is canonical because it is precisely the pullback of the task geometry on function space to parameter space:

$$M_T = \Psi^* g_T.$$

Its rank gives an exact but coarse local dimension, and its effective rank gives a practical continuous relaxation. Here $\text{erank}(M) := \exp(H(\hat{\sigma}))$ where $\hat{\sigma}$ is the normalized spectrum of M and H is Shannon entropy; equivalently erank equals the exponential of the spectral entropy, providing a smooth interpolant between rank counting and trace-to-operator-norm ratio. The key point is conceptual: the object that links architecture to task is not a scalar alone, but a task-weighted quadratic form on parameter space.

Remark 5.4 (Coordinate dependence). The pullback metric $M_T = J_\star^\top G_T J_\star$ depends on the chosen parameterization: a smooth reparameterization $\theta \mapsto \phi(\theta)$ replaces M_T by a congruent matrix, and erank is not congruence-invariant. All erank-based quantities in this paper (k_{use}, τ_{rep}, γ_{cov}) are therefore defined relative to the canonical parameterization of the architecture. This is the standard convention whenever effective rank is used as a diagnostic (cf. stable rank, participation ratio), and it is consistent with the fact that ECC itself is parameterization-independent while utilization inherently depends on the coordinate system in which optimization takes place.

5.4 Operational representational efficiency

The most important operational definition in this paper is the following.

Definition 5.5 (Usable task-aligned dimension)**.** *For architecture A and dataset or task family D, define*

$$k_{\text{use}}(A, D) := \sup_{\theta} \operatorname{erank}\big(M_{A,D}(\theta)\big).$$

For a specific checkpoint $\theta^\star$, the local *or* realized *usable dimension is*

$$k_{\text{use}}^{\text{loc}}(\theta^\star, D) := \operatorname{erank}\big(M_{A,D}(\theta^\star)\big) \;\le\; k_{\text{use}}(A, D).$$

The architecture-level quantity k_{use} measures what the architecture can in principle make usable; the checkpoint-level quantity $k_{\text{use}}^{\text{loc}}$ measures what a particular trained model currently uses.

Definition 5.6 (Representational efficiency and coverage)**.** *Let $C_A = \mathrm{ECC}(A)$ and let $k_D = \operatorname{erank}(G_D)$ denote the effective dimension of the task family. Define*

$$\boxed{\eta_{\text{eff}}(A, D) = \frac{k_{\text{use}}(A, D)}{C_A}, \qquad \gamma_{\text{cov}}(A, D) = \frac{k_{\text{use}}(A, D)}{k_D}.}$$

We identify

$$\boxed{\tau_{\text{rep}} := \eta_{\text{eff}}.}$$

Remark 5.7 (Computability). Given a trained parameter $\theta^\star$, evaluate $J_\star = D\Psi(\theta^\star)$ by automatic differentiation and the local task metric G_T from the specified loss at $f^\star = \Psi(\theta^\star)$ (I for squared loss, Fisher for cross-entropy, or $D^2 L_T(f^\star)$ for any smooth loss). Then

$$M_{A,D}(\theta^\star) = M_T(\theta^\star) = J_\star^\top G_T J_\star$$

is explicitly determined, and

$$\operatorname{erank}(M_T(\theta^\star)) = \exp(H(\tilde{\lambda})), \qquad \tilde{\lambda}_i = \lambda_i / \sum_j \lambda_j,$$

is computed from its spectrum. This is the operational usable dimension at $\theta^\star$; because Definition 5.6 takes $\sup_\theta$, it gives a computable lower bound on $k_{\text{use}}(A, D)$. Moreover, $C_A = \mathrm{ECC}(A)$ is architecture-determined and $k_D = \operatorname{erank}(G_D)$ is computed directly from the task-family metric, so η_{eff} and γ_{cov} follow immediately. For large models, one uses only matrix-vector products $v \mapsto J_\star^\top(G_T(J_\star v))$; Hutchinson trace estimators and randomized SVD/Lanczos then provide polynomial-time approximations without forming M_T.

The identity

$$\boxed{\mathrm{ECC} \cdot \tau_{\text{rep}} = k_{\text{use}}}$$

shows that representational alignment is precisely the fraction of architectural capacity that is genuinely usable for the task. Coverage and efficiency are different:

- High γ_{cov} means the architecture can touch most task-relevant directions.
- High η_{eff} means it does so without wasting much capacity on irrelevant directions.

This distinction clarifies the universal-versus-specialized trade-off. Broad architectures can have high coverage but mediocre efficiency. Specialist architectures can achieve high efficiency on matched domains but lose coverage outside them.

Remark 5.8 (Mixture-of-Experts routing as representational efficiency). Mixture-of-Experts provides a clean illustration. An MoE layer may have very large total ECC because it contains many experts, but task performance depends on how much of that capacity the router makes genuinely usable. In the present framework, the relevant object is the task pullback metric M_T of the combined router–expert system: better routing increases the task-aligned usable dimension relative to total ECC. Poor routing leaves many expert directions dormant or redundant, so $\tau_{\text{rep}} = k_{\text{use}}/\,\text{ECC}$ remains small even when the expert pool is large. Good routing matches different input regimes to experts whose tangent directions align with task geometry, increasing τ_{rep}. For a fixed routing architecture, failures to *learn* such a gate may additionally appear as low τ_{opt}; the representational point concerns the usable fraction that the routing scheme makes available in principle.

5.5 Self-MLE, recursive improvement, and bounds

The self-training fixed-point result (§7) showed that self-training without external information is a fixed point under data processing. The intelligence framework sharpens the conclusion. If the architecture is unchanged and no external channel is added, then one cannot increase τ_{stat} beyond the information already present. However, architecture modification can change ECC or τ_{rep}, and external feedback can change τ_{stat} or τ_{opt}. Recursive improvement is therefore not impossible in principle, but it is bounded and structurally decomposable rather than magically unbounded.

5.6 Practical levers

The three factors in the master equation correspond to three distinct engineering levers:

Intervention	Primary effect	Interpretation
Scale width/depth	ECC $\uparrow$	raises the ceiling
Architecture redesign	$\tau_{\text{rep}} \uparrow$ (and possibly ECC $\uparrow$)	improves task alignment
Better data	$\tau_{\text{stat}} \uparrow$	identifies more aligned directions
Better optimization	$\tau_{\text{opt}} \uparrow$	reaches more of what data supports
LoRA / fine-tuning / RLHF	mostly $\tau_{\text{rep}} \uparrow$	reallocates usable directions
Distillation / pruning	ECC $\downarrow$ with target τ preserved	compresses the useful subspace

6 A Conditional Coding Theorem and the Bridge Problem

6.1 A conditional coding theorem

Fix a task T with target function $f^\star$, and let $\mathcal{F}_{T,k}$ denote a k-dimensional aligned submodel. The following result formalizes the standard approximation-estimation-optimization decomposition.

Theorem 6.1 (Conditional coding theorem)**.** *Assume:*

(1) n i.i.d. samples are drawn from a distribution supported on a compact d_{data}-dimensional manifold, the squared loss is bounded by B^2, and $f^\star$ belongs to a β-smooth class;

(2) the aligned submodel $\mathcal{F}_{T,k}$ admits approximation rate

$$A_T(k) := \inf_{f \in \mathcal{F}_{T,k}} \|f - f^\star\|_{L^2}^2 \le C_{\text{app}} k^{-2\beta/d_{\text{data}}};$$

(3) $\log N(\varepsilon, \mathcal{F}_{T,k}, \|\cdot\|_\infty) \le c_0 k \log(c_1/\varepsilon)$;

(4) empirical risk minimization is solved to optimization error δ_{opt}.

Then the learned predictor $\hat{f}_{k,n}$ obeys

$$\mathbb{E}\|\hat{f}_{k,n} - f^\star\|_{L^2}^2 \lesssim C_1 k^{-2\beta/d_{\text{data}}} + C_2 \frac{k \log n}{n} + \delta_{\text{opt}}.$$

Remark 6.2. The theorem is called *conditional* because the approximation term is assumed, not derived from ECC alone. The whole difficulty of scaling-law theory is concentrated in that bridge.

6.2 Multiplicative utilization versus additive risk

The multiplicative τ decomposition and the additive excess-risk decomposition live at different levels:

$$\text{Intelligence}(T) = \text{ECC} \cdot \tau_{\text{rep}} \tau_{\text{stat}} \tau_{\text{opt}}$$

versus

$$\text{excess risk} = \text{approximation} + \text{generalization} + \text{optimization}.$$

The factors $\tau_{\text{rep}}, \tau_{\text{stat}}, \tau_{\text{opt}}$ should be understood as the quantities that determine the effective arguments of the additive bound: aligned dimension k, effective sample size, and optimization gap.

6.3 An entropy bridge

Theorem 6.3 (Entropy bridge)**.** *Let V_T be a k-dimensional parameter subspace and consider*

$$\mathcal{F}_{T,k} = \Psi\big(\theta^\star + (B_r \cap V_T)\big).$$

If Ψ is C^1 and L-Lipschitz on $\theta^\star + B_r$, then

$$\log N(\varepsilon, \mathcal{F}_{T,k}, \|\cdot\|_\infty) \le k \log\Big(1 + \frac{2Lr}{\varepsilon}\Big).$$

Proof. The parameter set $B_r \cap V_T$ is a k-dimensional Euclidean ball. Its covering number at radius δ is bounded by $(1 + 2r/\delta)^k$. Lipschitz continuity of Ψ transfers a δ-cover in parameter space into an $L\delta$-cover in function space. Setting $\delta = \varepsilon/L$ yields the claim. □

6.4 Why a purely static bridge fails

The hardest question is whether ECC, perhaps together with generic task regularity, can predict approximation quality by itself. Four counterexamples show that the answer is no.

Proposition 6.4 (Rank does not imply approximation)**.** *There exists an analytic full-rank family whose functions cannot approximate a simple target.*

Proof. Consider

$$\Psi(\theta)(x) = \sum_{j=1}^{P} \theta_j x^j, \qquad \theta \in [-1,1]^P, \quad x \in [0,1].$$

The Jacobian has full rank P, but every realized function vanishes at $x = 0$. Hence the constant function 1 cannot be approximated uniformly. □

Proposition 6.5 (Global universal approximation does not imply local aligned approximation)**.** *Even if an architecture family is globally universal in the uniform norm on a compact input domain, a fixed local slice around a trained point need not contain good uniform approximants (hence need not contain good L^2-approximants either).*

Reason. Universal approximation is a global existence statement: somewhere in parameter space there is a good solution in the chosen approximation norm. The bridge required for aligned submodels is local: inside a particular low-dimensional slice near a trained point there must be a good solution in that same norm. These are different claims. Compact parameter domains make the mismatch especially clear, because continuous images of compact domains cannot be dense in infinite-dimensional Banach spaces such as $C(\mathcal{X})$ equipped with the uniform norm. □

Proposition 6.6 (M_T-optimal slices need not be approximation-optimal)**.** *Even the top eigenspace of the pullback task metric may be worse for approximation than another slice of the same dimension.*

Proof. Take the linearized model with

$$J = \mathrm{diag}(10, 1), \qquad G_T = I, \qquad M_T = J^\top G_T J = \mathrm{diag}(100, 1).$$

The top-1 eigenspace is $\mathrm{span}(e_1)$, but if the residual target is e_2 then approximation along $\mathrm{span}(e_1)$ incurs error 1, whereas approximation along $\mathrm{span}(e_2)$ attains zero error. Thus the variance-maximizing direction need not be the bias-reducing direction. □

Example 6.7 (Dormant-neuron trap)*.* In a one-hidden-layer universal approximator

$$\Psi(\theta)(x) = \sum_i a_i \sigma(w_i \cdot x + b_i),$$

a neuron with tiny outer weight $a_i \approx 0$ may still contain crucial inner-weight structure for the task. But

$$\frac{\partial \Psi}{\partial (w_i, b_i)} \propto a_i$$

is then almost invisible to the Jacobian and therefore almost invisible to M_T. An aligned local slice can freeze the very neuron whose nonlinear relocation would be needed for optimal approximation. In this regime the model collapses toward a random-feature approximation class, which is much weaker than the full nonlinear network class.

Together these examples show that the static bridge from ECC to task performance is not merely unknown; it is false in general.

6.5 The correct local bridge: residual tails of the pullback task metric

The right local object is not a universal approximation law but a measurable residual-tail quantity.

Definition 6.8 (Pullback task metric)**.** *Let $\theta^\star$ be a trained point, $J_\star = D\Psi_{\theta^\star}$, and let G_T be a task Hessian, Fisher matrix, or Gauss-Newton metric on function space. The pullback task metric is*

$$\boxed{M_T(\theta^\star) := J_\star^\top G_T J_\star.}$$

Let $V_{T,k}$ denote the top-k eigenspace of M_T, and let $e_T = f_T^\star - \Psi(\theta^\star)$ be the residual at $\theta^\star$.

Definition 6.9 (Aligned residual tail)**.** *For a radius $r_k > 0$, define*

$$\eta_T(k) := \inf_{\delta \in V_{T,k},\, \|\delta\| \le r_k} \|e_T - J_\star \delta\|_{G_T}.$$

Theorem 6.10 (Local aligned approximation bridge)**.** *Assume the task loss is locally quadratic,*

$$L_T(f) = \frac{1}{2}\|f - f_T^\star\|_{G_T}^2,$$

and that Ψ is C^2 on $\theta^\star + B_{r_k}$ with second-order bound $\|R(\delta)\|_{G_T} \le \frac{1}{2}M_2\|\delta\|^2$ in its Taylor remainder

$$\Psi(\theta^\star + \delta) = \Psi(\theta^\star) + J_\star\delta + R(\delta).$$

Then the aligned local approximation error

$$A_T^{\mathrm{loc}}(k) := \inf_{\delta \in V_{T,k},\, \|\delta\| \le r_k} L_T\big(\Psi(\theta^\star + \delta)\big) - L_T^\star$$

satisfies

$$\frac{1}{2}\Big(\eta_T(k) - \frac{1}{2}M_2 r_k^2\Big)_+^2 \le A_T^{\mathrm{loc}}(k) \le \frac{1}{2}\Big(\eta_T(k) + \frac{1}{2}M_2 r_k^2\Big)^2.$$

Proof. For any admissible δ,

$$f_T^\star - \Psi(\theta^\star + \delta) = e_T - J_\star\delta - R(\delta).$$

Taking the G_T-norm and using the triangle inequality gives

$$\|e_T - J_\star\delta\|_{G_T} - \|R(\delta)\|_{G_T} \le \|f_T^\star - \Psi(\theta^\star + \delta)\|_{G_T} \le \|e_T - J_\star\delta\|_{G_T} + \|R(\delta)\|_{G_T}.$$

The remainder bound yields

$$\|e_T - J_\star\delta\|_{G_T} - \frac{1}{2}M_2 r_k^2 \le \|f_T^\star - \Psi(\theta^\star + \delta)\|_{G_T} \le \|e_T - J_\star\delta\|_{G_T} + \frac{1}{2}M_2 r_k^2.$$

Since norms are nonnegative, the left-hand side is at least $(\|e_T - J_\star\delta\|_{G_T} - \frac{1}{2}M_2 r_k^2)_+$. Squaring, multiplying by $1/2$, and taking the infimum over $\delta \in V_{T,k} \cap B_{r_k}$ proves the claim. □

If $G_T^{1/2}J_\star = U\Sigma V^\top$ and r_k is large enough that the unconstrained least-squares minimizer lies in B_{r_k}, then

$$\eta_T(k)^2 = \sum_{i>k} |\langle G_T^{1/2}e_T, u_i\rangle|^2 + \|e_{T,\perp}\|^2,$$

where $e_{T,\perp}$ denotes the component of $G_T^{1/2}e_T$ orthogonal to $\mathrm{im}(G_T^{1/2}J_\star)$. Thus $\eta_T(k)$ is exactly the tail of the residual in the task-singular basis of the trained model. Unlike $A_T(k)$, it is directly measurable from a trained model.

6.6 Why post-hoc $\eta_T(k)$ is still non-tautological and useful

The objection to any scaling theorem is tautology: does one need to solve the original approximation problem in order to verify the hypothesis? The local bridge avoids that problem. To compute $\eta_T(k)$ one needs $J_\star$, G_T, and the residual e_T, all of which are measurable after training. One does *not* need to search over all k-dimensional nonlinear submodels of the architecture. Thus $\eta_T(k)$ is not a rephrasing of the original task; it is a derived diagnostic.

Of course $\eta_T(k)$ is a post-training quantity, and we do not present it as an unconditional pre-training predictor. Its role is different: it is a *local diagnostic* of a realized model that can be estimated without re-solving the original nonlinear approximation problem. Channel capacity C likewise requires access to the channel law, yet C is not a mere restatement of the communication problem, because one need not search over all possible codes each time one evaluates it. Here, once a checkpoint $\theta^\star$ is given, $\eta_T(k)$ is obtained from $J_\star$, G_T, and the residual (or their empirical estimates on held-out task data) by linear-algebraic operations only. This makes it useful for: comparing checkpoints or architectures at equal training budget, choosing compression rank k, and identifying which task-singular directions remain unrecovered before deciding where further fine-tuning effort should be spent.

6.7 From local to dynamic: a six-term decomposition

The local bridge is correct but intentionally narrow. Real training is dynamic and may move far from the original tangent space. The right global object is the optimizer-dependent reachable bias

$$A^{\mathrm{reach}}_{T,\mathcal{A}}(B) := \inf_{\theta \in \mathcal{R}_{\mathcal{A}}(B)} \big(L_T(\Psi(\theta)) - L^\star_T\big),$$

where $\mathcal{R}_{\mathcal{A}}(B)$ is the set of parameters reachable by optimizer class $\mathcal{A}$ under budget B.

A useful *structural decomposition* is obtained by fixing a concrete k-dimensional pipeline: choose the top-k spectral slice $V_{T,k}$, train within that slice with optimizer class $\mathcal{A}$ under budget B on n samples, and denote the resulting predictor by $\hat{f}^{B,n}_{T,k}$. Define the resulting total k-slice excess risk by

$$\mathcal{U}_T(k) := L_T(\hat{f}^{B,n}_{T,k}) - L^\star_T.$$

With this convention we write

$$\boxed{\mathcal{U}_T(k) \overset{\text{struct}}{=} \beta_{\mathrm{exist}}(k) + \Gamma_{\mathrm{curv}}(k) + \Gamma_{\mathrm{sel}}(k) + \Gamma_{\mathrm{act}}(B,k) + C_{\mathrm{gen}}\frac{k}{n} + \delta_{\mathrm{opt}}.} \tag{1}$$

Here $\overset{\text{struct}}{=}$ means "decomposed into the following six sources of gap" rather than a theorem-level identity between previously defined benchmark quantities.

The six terms are understood operationally as follows:

- $\beta_{\mathrm{exist}}(k)$ is the best global k-dimensional approximation error over admissible nonlinear submodels;
- $\Gamma_{\mathrm{curv}}(k)$ is the extra error incurred when one passes from the best global k-family to the best *local* k-family near the checkpoint;
- $\Gamma_{\mathrm{sel}}(k)$ is the extra error incurred by the particular slice-selection rule (here: top-k eigenspace of M_T) relative to the locally approximation-optimal k-slice;
- $\Gamma_{\mathrm{act}}(B,k)$ is the reachability/dynamics gap between the best point in the chosen slice and the best point actually reachable by optimizer class $\mathcal{A}$ under budget B;
- $C_{\mathrm{gen}}k/n$ is shorthand for the statistical generalization term; if one wants the entropy factor explicit, read this as $C_{\mathrm{gen}}(k\log n)/n$, equivalently C_{gen} is allowed to absorb polylogarithmic factors in n;
- δ_{opt} is the residual solver-level suboptimality after the reachable set has been fixed. Thus $\Gamma_{\mathrm{act}}(B,k)$ is the optimizer-class/budget limitation of the training dynamics, whereas δ_{opt} is the remaining numerical gap of the specific finite optimization run inside that reachable regime.

The four negative results above correspond almost one-to-one to the first four terms. Rank alone does not control β_{exist}. Global universality does not kill Γ_{curv}. M_T-optimality does not kill Γ_{sel}. Optimization traps and dormant neurons live in Γ_{act}.

Remark 6.11 (ECC is ceiling, not predictor). The role of ECC is now clear. It bounds the ambient range of k:

$$k \le \mathrm{ECC}\,.$$

It is therefore a resource variable or ceiling, analogous to channel capacity in information theory. It is not, by itself, a standalone performance predictor.

7 Framework in Action

7.1 Capacity geometry explains architectural phenomena

7.1.1 Bilinear attention is Pareto-optimal on the score side

Reference [1] showed that bilinear attention saturates the natural d^2 score-side ceiling of degree-2 interactions. The present factorized result preserves that conclusion at the architectural level: among degree-at-most-two score families with comparable evaluation cost, bilinear score computation remains on the Pareto frontier of score-side ECC versus compute.

A useful summary is:

$$\text{bilinear score: ECC} \sim d^2, \qquad \text{compute} \sim O(nd^2),$$

whereas additive score families have only $O(d)$ score-side capacity, and more complicated degree-2 families do not improve the asymptotic ECC/compute trade-off. The value-output correction of Theorem 4.4 changes the full attention count, but it does not remove the optimality of the bilinear score mechanism itself.

7.1.2 Sparse attention when $d^2 \ll n^2$

A standard attention layer compares n^2 token pairs, but the score-side functional freedom scales like d^2 (or d_h^2 per free head kernel, and $d_h(2d - d_h)$ after factorization). When $n \gg d$, most pairwise token relations cannot be independently controlled.

This yields a simple geometric interpretation of sparse attention: sparse patterns are not discarding a large pre-existing pairwise capacity. Rather, they acknowledge that the controllable score geometry already lives in a much smaller d^2-scale family. For a single head with $d_h = 128$ and context length $n = 131072$,

$$\frac{d_h^2}{n^2} \approx 10^{-6},$$

so a dense $n \times n$ score matrix is a massively overcomplete carrier relative to score-side functional dimension.

7.1.3 LoRA as accessible low-dimensional geometry

Consider a rank-r LoRA adapter on an $m \times n$ weight matrix. The adapter family is not r-dimensional; it is the rank-r matrix variety, whose dimension is

$$r(m + n - r).$$

Thus the task-accessible dimension under LoRA obeys

$$k_{\text{LoRA}}(T, \theta^*) \leq \min\Big(\text{ECC}, \sum_\ell r_\ell(m_\ell + n_\ell - r_\ell)\Big).$$

For a square $d_h \times d_h$ per-head adapter,

$$k_h \leq \min(d_h^2,\ r(2d_h - r)).$$

This correction matters. A rank-8 update on a 4096×4096 matrix accesses roughly 6.5×10^4 directions, not 8 directions. ECC still supplies an absolute ceiling, but the geometry of low-rank varieties explains why LoRA can be far more expressive than naive rank counting suggests.

7.1.4 Catastrophic forgetting as subspace geometry

A simple capacity-overflow criterion such as $k_1 + k_2 > \text{ECC}$ is neither necessary nor sufficient for forgetting. The correct object is the geometry of task-active subspaces U_1 and U_2.

Their joint dimensional demand is

$$\dim(U_1 + U_2) = \dim U_1 + \dim U_2 - \dim(U_1 \cap U_2).$$

Forgetting pressure is high when either:

1. the joint subspace exceeds local usable capacity, or
2. optimization directions beneficial to task 2 are harmful to task 1, even when the subspace dimensions would fit.

Thus catastrophic forgetting is governed by subspace overlap, principal angles, and optimization conflict, not just by the scalar sum of task dimensions.

7.2 Intelligence decomposition explains training dynamics

7.2.1 Self-training as a fixed point

Let $Y^\star$ denote latent ground truth, D_{orig} the original dataset, and D_{syn} synthetic data sampled from a model trained only on D_{orig} without external information. Then the Markov structure

$$Y^\star \to D_{\text{orig}} \to \theta \to D_{\text{syn}}$$

implies, by the data-processing inequality,

$$I(D_{\text{syn}}; Y^\star) \le I(\theta; Y^\star) \le I(D_{\text{orig}}; Y^\star).$$

Therefore self-MLE with the same architecture and no external information cannot improve the statistical factor τ_{stat}. Synthetic data only help when they inject outside information: stronger teachers, environment feedback, human preference feedback, verifiers, or architecture changes.

7.2.2 Architecture screening before training

ECC and ECC/compute can be estimated without running a full pretraining experiment. This suggests a practical screening principle: before spending large-scale compute, compare candidate architectures by

$$(\text{ECC},\ \text{ECC}/\text{FLOP},\ \text{expected } \tau_{\text{rep}} \text{ for the target workload}).$$

The first quantity measures capacity ceiling, the second measures cost efficiency, and the third measures likely alignment to the task family. This is not a complete performance predictor, but it is a useful first-pass filter.

7.2.3 Emergence as threshold crossing

Suppose a task T requires effective aligned dimension $k(T)$. Let $k_{\text{eff}}(\theta)$ denote the realized task-aligned dimension after training. Then:

$$k_{\text{eff}}(\theta) < k(T) \ \Rightarrow\ \text{task failure}, \qquad k_{\text{eff}}(\theta) \gtrsim k(T) \ \Rightarrow\ \text{can produce rapid performance gain.}$$

If scaling smoothly increases ECC and slowly improves alignment, then $k_{\text{eff}}(\theta)$ grows smoothly while evaluation metrics such as accuracy exhibit an abrupt threshold effect. On this view, many "emergent" abilities are threshold crossings of a smooth underlying quantity rather than discontinuous ontological jumps.

7.2.4 Capacity versus utilization

The same model can be spectacular on one task and brittle on another because ECC is architecture-wide while utilization is task-specific:

$$\text{Intelligence}_T = \text{ECC} \cdot \bar{\tau}_T.$$

Pretraining, fine-tuning, RLHF, and LoRA mostly change τ rather than ECC. Distillation often lowers ECC while trying to preserve task-aligned directions. This gives a unified interpretation of common training interventions: most practical progress is not the creation of new first-order capacity but the redistribution or exploitation of existing capacity.

7.2.5 Workload-aware compression and data curation

Given a deployment workload $W = \{(T_i, w_i)\}_{i=1}^m$, define the workload metric

$$M_W := \sum_{i=1}^{m} w_i M_{T_i}.$$

Compression should preserve the leading eigenspace of M_W, because those are the directions the workload actually rewards. If u_j are the singular directions of the workload-aligned Jacobian, then the compression-induced degradation is controlled by a tail quantity of the form

$$\eta_W(k)^2 \approx \sum_{j>k} |\langle G_W^{1/2} e_W, u_j \rangle|^2.$$

The same object has a dual pretraining use. Instead of asking which directions may be discarded after training, one may ask which directions should be supplied before training. A pilot model can estimate a target workload metric M_W, and candidate data sources can be ranked by how strongly they excite its dominant eigenspaces. In this sense, workload-aware data curation is the dual of workload-aware compression.

7.3 Interpretive scope and limits

7.3.1 Separating the technical tension in the parrot and scaling debates

The familiar claims

"scaling works" and "scaling wastes structure"

are not contradictory once intelligence is decomposed as

$$\text{Intelligence}(T) = \text{ECC}\, \tau_{\text{rep}} \tau_{\text{stat}} \tau_{\text{opt}}.$$

Increasing scale raises ECC, so performance can improve dramatically. But if $\tau_{\text{rep}} < 1$, the architecture is paying a multiplicative tax for using a representational form that is not well matched to the task family. Likewise, out-of-distribution failures are naturally associated with low representational alignment, while strong in-distribution performance can come from high statistical efficiency even when representational efficiency is imperfect.

The framework therefore resolves a *technical tension*: some arguments are about coverage through high ECC, others about efficiency through high τ_{rep}. It does *not* resolve the full philosophical debate about meaning, grounding, or semantics.

7.3.2 Limits of the framework

The framework is intentionally geometric and functional. It does not answer questions about sentience, subjective experience, deceptive alignment, or metaphysical "true understanding." It also does not produce exact societal risk thresholds, market forecasts, or policy advice. Its correct scope is narrower and more technical: a principled capacity ceiling, measurable local alignment objects, and a decomposition of performance into architectural, statistical, and optimization factors.

8 Future Directions in Architectural Design

8.1 Complex-valued and quaternionic attention

If the attention score kernel lives over $\mathbb{C}$ rather than $\mathbb{R}$, then a complex bilinear form carries twice the real dimension.

Proposition 8.1 (Complex-valued attention)**.** *For an unconstrained complex bilinear score kernel on a d_h-dimensional complex head space,*

$$\boxed{\mathrm{ECC}_{\mathbb{C}} = 2d_h^2}$$

as a real dimension count.

More generally, for the real division algebras $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$, the score-side ceiling scales as

$$\mathrm{ECC}_{\mathbb{A}} = (\dim_{\mathbb{R}} \mathbb{A})\, d_h^2 \qquad \text{for } \mathbb{A} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}.$$

This does *not* imply that every practical implementation gains that full factor. If a complex representation is immediately projected back to a real score space before softmax, the gain can collapse. In practice the main advantage may come less from raw ECC multiplication than from a better inductive bias and therefore higher τ_{rep} on matched domains.

8.2 Beyond degree 2: from direct sums to tensor products

Current multi-head attention aggregates heads primarily by direct sum. If head interactions were composed by tensor-product-like couplings rather than mere concatenation, one might hope for multiplicative rather than additive functional growth across subsystems. This is still speculative, but it suggests a program: instead of asking whether attention should be "higher degree" in the abstract, ask whether head composition should move from

$$\oplus \quad \text{toward} \quad \otimes\,.$$

Any successful version of that idea would amount to engineering interactions between the currently separate functional subspaces of different heads.

8.3 The data wall and synthetic data

If the original data sources are exhausted, then τ_{stat} approaches a ceiling. The self-training analysis (§7) implies that self-generated data cannot break that ceiling in the absence of external information. Thus the routes forward are limited: new data sources, environment interaction, stronger external teachers or verifiers, or architecture changes that increase τ_{rep} so the same information is used more efficiently.

8.4 A Shannon-style roadmap

Information theory reached its constructive high points only decades after Shannon identified capacity. The same may happen in AI.

Coding-theory milestone	AI analogue	Status
Shannon capacity	ECC as capacity ceiling	established here
Rate-distortion tools	τ decomposition and $\eta_T(k)$ diagnostics	established here
Turbo / LDPC era	optimization methods driving $\tau_{\text{opt}} \to 1$	open
Constructive polar coding	architectures with $\tau_{\text{rep}} \approx 1$ by design	grand challenge

This analogy is heuristic, but it is productive. Without ECC and the τ decomposition one cannot even formulate the analogue of "capacity-approaching codes" for AI architectures. With them, the question becomes mathematically expressible: can one construct architecture families whose representational efficiency approaches one on a specified task family?

8.5 No universal representational optimum

To discuss universal versus specialized architectures, we first need a mismatch proxy.

Definition 8.2 (Algebraic efficiency proxy)**.** *Let $L_A \subset \mathrm{End}_{\mathbb{R}}(V)$ be the admissible linear maps of architecture A, and let $S_D \subset \mathrm{End}_{\mathbb{R}}(V)$ be the task-relevant subspace for domain D. Orthogonal projectors P_{L_A}, P_{S_D} are defined with respect to the Frobenius inner product on $\mathrm{End}_{\mathbb{R}}(V)$. Define*

$$\tau_{\text{rep}}^{\text{lin}}(A, D) = \frac{\mathrm{tr}(P_{L_A} P_{S_D})}{\dim L_A}.$$

This quantity measures how much of the architecture's admissible linear family overlaps with the relevant domain structure. For example, in a complex-structured domain one may distinguish J-commuting from J-anticommuting directions. A complex architecture may achieve near-unit efficiency on one and near-zero efficiency on the other, while a real architecture covers both only diffusely.

Theorem 8.3 (No universal τ_{rep} optimum)**.** *Suppose there exist architectures A_1, A_2 and domains D_1, D_2 such that*

$$\tau_{\text{rep}}(A_1, D_1) > \tau_{\text{rep}}(A_2, D_1), \qquad \tau_{\text{rep}}(A_1, D_2) < \tau_{\text{rep}}(A_2, D_2),$$

and neither architecture dominates the other on both domains simultaneously. Then no single architecture in $\{A_1, A_2\}$ maximizes expected representational efficiency for every probability distribution supported on $\{D_1, D_2\}$.

Proof. For a mixture weight $p \in [0, 1]$, the expected efficiency is

$$E_i(p) = p\, \tau_{\text{rep}}(A_i, D_1) + (1 - p)\, \tau_{\text{rep}}(A_i, D_2).$$

By hypothesis, $E_1(1) > E_2(1)$ and $E_1(0) < E_2(0)$. Hence the affine functions E_1 and E_2 cross, so the maximizing architecture between A_1 and A_2 depends on p. The non-domination hypothesis ensures neither is uniformly superior. □

Remark 8.4. This is related in spirit to no-free-lunch ideas, but it is not the same theorem. The statement is resource-theoretic and structured: once mismatch crossovers exist, universality and efficiency cannot be simultaneously optimized everywhere.

8.6 Tokenization as structural projection

Modern multimodal systems often reduce diverse domains to a common token space before sending them into a single backbone. In our framework, this is a representational projection. The price of universality is that native domain structure can be flattened:

$$\tau_{\text{rep}}(\text{tokenized domain}) < \tau_{\text{rep}}(\text{native structured domain})$$

whenever important symmetries, topologies, phases, or geometric relations are lost in the projection. Scaling remains a valid way to compensate, because increasing ECC still increases intelligence. But it does so inefficiently when τ_{rep} is low.

9 Numerical Illustrations

To illustrate that the mathematical objects defined in this paper—eigenvalue spectra, aligned slices, and the local bridge residual $\eta_T(k)$—make contact with computation on real models, we report three small-scale experiments. Their purpose is not to claim engineering improvements but to check that these objects are computable and carry empirically detectable signal. All experiments use micro-scale models and subspaces because full Fisher eigendecomposition scales as $O(P^3)$; the claim is not that these scales are practically important but that the theoretical quantities are measurable. Full reproducible code and raw outputs are provided in the supplementary `experiment_package/`.

9.1 Experiment 1: α-crossover validation

1a: Algebraic sanity check. As a baseline consistency test, we construct a synthetic eigensystem ($n = 30$, $\lambda_i = i^{-2}$, planted $c_i(\alpha) = \lambda_i^{(1-2\alpha)/2}$) and verify that the top-k and $|c_i|^2$-guided selectors coincide when $\alpha < \frac{1}{2}$ and diverge when $\alpha > \frac{1}{2}$, with an exact crossover at $\alpha = \frac{1}{2}$. This is a deterministic algebraic identity and serves only to confirm that the implementation is correct.

1b: Teacher-student held-out prediction. To test whether $\hat{\alpha}$ estimation has predictive power beyond the algebraic identity above, we use a teacher-student protocol on the `digits` dataset with a three-layer MLP ($d_{\text{hidden}} = 32$, $P \approx 2500$ parameters, $k = 10$ aligned directions). For each trial: (i) sample $\alpha_{\text{true}} \in [-0.2, 1.2] \setminus (0.45, 0.55)$ (the neutral band is excluded so the binary prediction is margin-separated); (ii) construct a unit-norm teacher perturbation $\delta_{\text{teacher}} = W \cdot (\lambda^{-\alpha_{\text{true}}} \odot \hat{g})/\|\cdot\|$ satisfying $\mathrm{SC}(\alpha, g)$ by construction; (iii) estimate $\hat{\alpha}$ from a *separate* held-out gradient (disjoint from the gradient used for the teacher) via log-log regression of $|\hat{c}_i|^2$ versus λ_i; (iv) predict top-k ($\hat{\alpha} < \frac{1}{2}$) or bottom-$k$ ($\hat{\alpha} > \frac{1}{2}$); (v) solve least-squares interventions in each subspace on an adaptation set and evaluate on a strictly held-out test set. Data is split four ways (metric/alpha/adapt/test, 80 samples each) with no overlap.

Of 60 trials, all 60 returned a valid prediction (no ties or solver failures). The prediction accuracy is $43/60 = 0.717$ (95% CI: 0.59–0.83) with one-sided binomial $p = 5.33 \times 10^{-4}$. The sub-unit accuracy reflects the difficulty of log-log regression from noisy finite-sample gradients, not a failure of the planted mechanism. This confirms that, in a well-specified teacher-student setting where $\mathrm{SC}(\alpha, g)$ holds by construction, the $\hat{\alpha}$-based binary crossover prediction is statistically above chance on held-out data.

1c: GPT-2 held-out α-crossover. To verify that the prediction mechanism transfers to a production-scale model, we run the same held-out protocol on GPT-2 (layer-11 attention, ∼590K parameters). For each of 3 seeds × 3 out-of-distribution domains (Code, Korean, JSON), we estimate $\hat{\alpha}$ from a task-metric eigenbasis computed on a support set of 25 sequences, predict top-k or bottom-k ($k = 8$), fine-tune in each constrained subspace for 20 steps, and evaluate on a strictly held-out set. Data splits (metric/alpha/train/eval) are fully disjoint.

The analysis units are the full nine pre-specified domain-seed pairs

$$\{(\text{Code}, s), (\text{Korean}, s), (\text{JSON}, s) : s = 1, 2, 3\}.$$

Seven of these nine pairs satisfied the pre-registered quality filters

$$R^2 \geq 0.1, \qquad |\hat{\alpha} - 0.5| \geq 0.05,$$

so the scored-sample coverage is explicitly 7/9. On that retained sample, the $\hat{\alpha}$-based rule was correct in all 7/7 cases, giving one-sided binomial $p = 0.0078$. Every retained case lay in the $\hat{\alpha} < \frac{1}{2}$ regime, and in every such case top-k outperformed bottom-k. A trivial always-top-k predictor would therefore also achieve 7/7 on the retained sample.

This does not undermine the theoretical claim. The contribution of the theory is not a post-hoc observation that top-k happened to win in these runs; rather, the Gram-decomposition mechanism specifies *which*

measurable quantity to compute ($\hat{\alpha}$) and *which* threshold to test ($\alpha = \frac{1}{2}$) before looking at the adaptation outcome. The GPT-2 sample happens to be one-sided, so this experiment should not be read as establishing predictive value over an always-top-k rule on that retained subset. Its role is different: it confirms that the theoretical quantity $\hat{\alpha}$ is computable in a production model and that its predictions are consistent with observed adaptation outcomes. The non-trivial crossover claim itself—namely, that $\alpha = \frac{1}{2}$ is the correct boundary between the two selection regimes—is validated in Experiment 1b, which probes both sides of the boundary under controlled conditions.

9.2 Experiment 2: GPT-2 subspace intervention

Setup. We extract the Fisher eigenbasis for GPT-2's layer-11 attention weight (∼590K parameters) using English-language gradients and fine-tune in constrained $k = 10$ subspaces on target domains that differ from the pretraining distribution. Layer 11 and $k = 10$ were chosen for computational tractability. To avoid information leakage, we use a strict three-way split: a *support set* (for computing $|c_i|^2$), a *training set* (for subspace fine-tuning), and a held-out *evaluation set*. We compare four selectors: (A) Fisher top-k (task-agnostic), (B) $|c_i|^2$-guided (target-aware, from support set only), (C) random within the Fisher basis, and (D) random in the ambient parameter space. Selector (B) has an inherent advantage over (A) because it uses target-domain information; the question is whether the *magnitude* of this advantage is predictable from the theory's energy-gap quantity.

Results across 8 domains. We test on Code, Korean, JSON, Math, Poetry, Medical, Legal, and Cooking domains. Treating domains as independent units, the $|c_i|^2$-guided selector wins on 6 of 8 domains. A one-sided sign test at the domain level gives $p \approx 0.145$, which is not individually significant; the two losses occur in domains where the energy gap between top-k and guided selections is small (Medical: 6.3%, Cooking: 26.6%), consistent with noisy $|c_i|^2$ estimation from small support sets.

Correlational test. Because the win/loss outcome alone is statistically thin, the more informative test is whether the *magnitude* of the guided selector's advantage is associated with the theory's own quantities. We define the *energy gap* as the difference in captured $|c_i|^2$ energy between the guided and top-k selectors. The Spearman rank correlation between energy gap and the percentage improvement of guided over top-k across the $n = 8$ domains is

$$r_s = 0.786, \quad p = 0.021.$$

This small-sample correlation is consistent with the theory's prediction that the $|c_i|^2$ ranking determines the degree of subspace misalignment. We note two caveats: (i) the comparison is between a target-aware and a task-agnostic selector, so it does not rule out other target-aware methods; (ii) $n = 8$ makes the rank correlation fragile to individual outliers. Nonetheless, the direction and magnitude of the association are as the theory predicts.

9.3 Experiment 3: Local bridge bound verification

We verify the local bridge theorem (Theorem 6.10) by computing the spectral residual $\eta_T(k)$ for a one-hidden-layer tanh MLP ($d_{\text{in}} = 16$, hidden width 32, $d_{\text{out}} = 4$; tanh is C^2, so the theorem's smoothness assumption holds). The teacher-student regression uses squared loss ($G_T = I$); a separate cross-entropy variant ($G_T \neq I$) is also included in the supplementary code. For each seed, $\eta_T(k)$ is evaluated at $k \in \{0, 1, 2, 4, 8, 16, 32, 64\}$ aligned eigenvectors, and we compare:

1. The theorem's predicted upper bound on the constrained optimization gap, with M_2 estimated as $2 \times \max$ of the Hessian-remainder norm over 64 random perturbations within B_{r_k} (where r_k is the norm of the unconstrained least-squares step, scaled by factor 1.5).
2. The actual gap obtained by nonlinear constrained optimization (Adam, 250 steps, projected to B_{r_k}) in the corresponding subspace.

Results. Across 5 seeds $\times$ 8 values of $k = 40$ evaluations along correlated (k)-curves, $\eta_T(k)$ achieves 0/40 bound violations in the $G_T = I$ setting and 0/40 in $G_T \neq I$. Since the theorem is mathematically proved, the absence of violations serves as a consistency check rather than independent evidence; the informative quantity is tightness. The mean prediction error is $< 0.25\%$ in both settings. This illustrates that $\eta_T(k)$ is a practically tight spectral diagnostic in this local small-network regime.

10 Related Work and Discussion

10.1 Related work

The closest mathematical antecedents are the neuromanifold and functional-dimension literatures. Henry, Marchetti, and Kohn compute the dimension of function spaces for lightning self-attention using algebraic geometry [2]. Their quantity coincides with generic Jacobian rank in the models they analyze, but they do not treat full softmax attention, value-output factorization, or the broader intelligence decomposition developed here. Grigsby, Lindsey, and Rolnick's work on hidden symmetries and functional dimension in ReLU networks is likewise closely related to our use of generic Jacobian rank [3].

The scaling-law literature studies how performance varies with data and model size [5, 6, 7, 8]. Havrilla and Liao provide an especially relevant constructive approximation theory for transformers on low-dimensional data manifolds [4]. Our coding-theorem section is compatible with that program, but it emphasizes the distinction between existence of good global approximants and accessibility of good local aligned slices.

The low-rank adaptation and intrinsic-dimension literatures study trainability and low-dimensional fine-tuning [9, 10, 11]. Our perspective clarifies why these methods can work: the accessible low-rank manifold can already be large in function space, but its performance depends on alignment to task geometry.

Finally, classical approximation theory, especially the work of Barron and the nonlinear approximation literature, informs our treatment of the bridge problem [12]. The dormant-neuron and local-slice counterexamples show that those approximation regimes must be distinguished carefully from linearized or NTK-style aligned slices.

10.2 Discussion

Does scaling alone imply AGI?

Scaling raises ECC, so it genuinely raises the ceiling. In that sense, the strongest pro-scaling argument is correct. But if τ_{rep}, τ_{stat}, or τ_{opt} remain bounded away from one, gains can be slow or domain-specific. The right conclusion is not "scaling fails" but "scaling is one lever among several." The equation

$$\mathrm{ECC}_{\text{req}}(I^\star) = \frac{I^\star}{\tau_{\text{rep}}\tau_{\text{stat}}\tau_{\text{opt}}}$$

makes this explicit: poor utilization forces one to buy intelligence with much larger capacity.

Are large language models just parrots?

The framework turns a binary debate into a spectrum. High coverage with low efficiency means that a model can imitate broad distributions while wasting most of its capacity on irrelevant directions; this is the regime most naturally associated with "parroting." High efficiency means that much more of the capacity is aligned with the latent task structure. Most real systems lie between those extremes, and their position depends strongly on the task family.

Transformer versus Mamba

The corrected ECC counts sharpen, but do not settle, the Transformer-Mamba comparison. Transformers can have larger single-block ECC, especially once full attention and FFN contributions are counted, while Mamba remains much cheaper per context step and therefore stronger in ECC-per-compute on long sequences. The right question is not which architecture is universally better, but which task distribution places higher value on random access, pairwise comparison, recurrence, compression, or streaming.

Is data curation an art or a science?

If workload metrics can be estimated, then data curation becomes partially algorithmic. The relevant criterion is not raw data volume but excitation of the dominant eigenspaces of the target workload metric M_W. This does not remove all empirical uncertainty, but it turns a large part of data selection from tacit craft into a measurable alignment problem.

Can there be a universal world model?

A universal architecture can have high coverage. The no-universal-τ_{rep} theorem shows only that high coverage does not imply maximal efficiency on every domain mixture. This leaves a clear trade-off: universal backbones may amortize across tasks, while specialized modules may reach much higher efficiency on structured domains such as vision, audio phase, or physics.

Is higher ECC always better?

Higher ECC is better only ceteris paribus. ECC is a ceiling, not a guarantee. Two models with the same ECC can differ drastically if one has much better representational alignment, better data, or better optimization. Conversely, a smaller model with much higher τ_{rep} may outperform a larger but badly mismatched model on the task family that matters.

11 Conclusion

This paper develops an information-theoretic view of AI architecture in which the central capacity object is

$$\boxed{\mathrm{ECC}(\Psi) = \mathrm{rank}_{\mathrm{gen}}(D\Psi).}$$

The quantity is not an ad hoc proposal: it is the unique invariant satisfying equivalence-invariance, additivity, and normalization. It counts functional freedom rather than storage size, and it supports explicit calculations.

The corrected architectural results are striking. Full multi-head attention contributes

$$\boxed{\mathrm{ECC}_{\mathrm{Attn}} = H[d_h(2d - d_h) + d_v(2d - d_v)]},$$

ReLU feed-forward layers contribute

$$\boxed{\mathrm{ECC}_{\mathrm{FFN}} = m(2d - 1)},$$

and standard Mamba blocks scale as

$$\boxed{\mathrm{ECC}_{\mathrm{Mamba}} = \tfrac{1599}{256}\, d^2 + O(d) \approx 6.246\, d^2 + O(d)}$$

under default hyperparameters. For standard transformers, per-component ECC is therefore close to parameter count, and compositional ECC is at most their sum. Most component parameters are functionally active. The main waste is not static capacity but misalignment.

That observation motivates the second half of the paper. Task performance is governed by

$$\boxed{\text{Intelligence}(T) = \text{ECC} \cdot \tau_{\text{rep}} \tau_{\text{stat}} \tau_{\text{opt}} \cdot}$$

The representational factor has the operational meaning

$$\boxed{\tau_{\text{rep}} = \frac{k_{\text{use}}}{\text{ECC}}},$$

so the product $\text{ECC}\,\tau_{\text{rep}}$ is exactly the usable task-aligned dimension. The pullback task metric $M_T = J_\star^\top G_T J_\star$ is the canonical local object, and its residual-tail statistic $\eta_T(k)$ provides a measurable local bridge from geometry to approximation error. Beyond the local regime, a six-term decomposition separates existence, geometry, selection, reachability, generalization, and optimization.

Several questions remain open. The most important are:

1. tighter compositional bounds on $\dim(\mathcal{S}_1 + \cdots + \mathcal{S}_L)$ for specific deep architectures beyond the Signature Criterion;
2. a theory of the dynamic reachable bias $A^{\text{reach}}_{T,\mathcal{A}}(B)$;
3. empirical estimation of $\eta_T(k)$ in trained large models;
4. constructive design of architectures with τ_{rep} close to one on specified task families.

The slogan inherited from [1] remains the right one. That paper defined the ceiling. The present paper makes precise how much of the ceiling is used, why the unused part matters, and why alignment, not raw block capacity, is the deeper bottleneck.

A Proof of Theorem 4.4 (Full MHA ECC)

Lemma A.1 (Kernel exactness for full multi-head attention)**.** *Assume that the attention input domain contains a nonempty open set of two-token sequences $X = (x_1, x_2) \in (\mathbb{R}^d)^2$ (in particular, the sequence length satisfies $n \geq 2$). Let*

$$\Psi_{\text{Attn}} : \prod_{h=1}^{H} \big(\mathbb{R}^{d\times d_h} \times \mathbb{R}^{d\times d_h} \times \mathbb{R}^{d\times d_v} \times \mathbb{R}^{d_v\times d}\big) \to \mathcal{F}$$

be the realization map

$$\Psi_{\text{Attn}}(\{W_Q^h, W_K^h, W_V^h, O_h\}_{h=1}^H)(X) = \sum_{h=1}^{H} P_h(X)\, X\, W_V^h O_h, \qquad P_h(X) = \text{softmax}\Big(\frac{X W_Q^h (W_K^h)^\top X^\top}{\sqrt{d_h}}\Big).$$

At any parameter point such that every W_Q^h, W_K^h has full column rank and every W_V^h, O_h has rank d_v, the kernel of $D\Psi_{\text{Attn}}$ is exactly

$$\bigoplus_{h=1}^{H} \mathfrak{g}_h^{QK} \oplus \bigoplus_{h=1}^{H} \mathfrak{g}_h^{VO},$$

where

$$\mathfrak{g}_h^{QK} = \Big\{(W_Q^h A_h,\, -W_K^h A_h^\top,\, 0,\, 0) : A_h \in \mathbb{R}^{d_h\times d_h}\Big\},$$
$$\mathfrak{g}_h^{VO} = \Big\{(0,\, 0,\, W_V^h B_h,\, -B_h O_h) : B_h \in \mathbb{R}^{d_v\times d_v}\Big\}.$$

Hence

$$\dim\ker D\Psi_{\text{Attn}} = H(d_h^2 + d_v^2)$$

on a Euclidean-open dense subset of parameter space, and therefore

$$\text{rank}_{\text{gen}}(D\Psi_{\text{Attn}}) = H\big[d_h(2d - d_h) + d_v(2d - d_v)\big].$$

Proof. For each head define the reduced coordinates

$$M_h := W_Q^h (W_K^h)^\top, \qquad N_h := W_V^h O_h.$$

Write $q_h(W_Q^h, W_K^h) = M_h$ and $v_h(W_V^h, O_h) = N_h$. At the stated full-rank points, $\ker Dq_h$ is exactly the tangent space to the right $\mathrm{GL}(d_h)$-orbit $(W_Q^h, W_K^h) \mapsto (W_Q^h G,\ W_K^h G^{-\top})$, namely

$$\ker Dq_h = \Big\{ (W_Q^h A_h,\ -W_K^h A_h^\top) : A_h \in \mathbb{R}^{d_h \times d_h} \Big\},$$

and similarly $\ker Dv_h$ is the tangent space to the $\mathrm{GL}(d_v)$-orbit. Thus it remains to prove that the differential of the reduced map

$$\widetilde{\Psi}(\{M_h, N_h\}_{h=1}^H)(X) = \sum_{h=1}^H P_h(X; M_h)\, X\, N_h$$

is injective on a Euclidean-open dense subset of reduced parameter space.

Assume therefore that $D\widetilde{\Psi}_{(M,N)}(\{\delta M_h, \delta N_h\}_{h=1}^H) = 0$ as an element of function space. We must show $\delta M_h = \delta N_h = 0$ for every h.

For two tokens write $X = (x_1, x_2)$, and set $\rho(t) := 1/(1 + e^{-t})$. For each head h, define

$$\alpha_h(X) := \frac{x_1^\top M_h (x_1 - x_2)}{\sqrt{d_h}}, \qquad \beta_h(X) := \frac{x_2^\top M_h (x_1 - x_2)}{\sqrt{d_h}}.$$

Then the h-th head output on the two-token slice is

$$\psi_h(X) = \begin{bmatrix} x_2 + \rho(\alpha_h)(x_1 - x_2) \\ x_2 + \rho(\beta_h)(x_1 - x_2) \end{bmatrix} N_h.$$

Step 1: $\delta N_h = 0$ and the symmetric part of δM_h vanishes.

Fix $u \in \mathbb{R}^d$ and consider the slice $X_u(t) = (tu, 0)$. Define $\mu_h(u) := u^\top M_h u/\sqrt{d_h}$, $\delta\mu_h(u) := u^\top \delta M_h u/\sqrt{d_h}$, and $z_h(u) := \mu_h(u)^2$. On this slice the first variation has the form

$$\delta\psi_h(X_u(t)) = \begin{bmatrix} t\rho(t^2\mu_h(u))\, u\, \delta N_h + t^3 \rho'(t^2\mu_h(u))\, \delta\mu_h(u)\, u N_h \\ (t/2)\, u\, \delta N_h \end{bmatrix}.$$

Since $\sum_h \delta\psi_h(X_u(t)) = 0$, subtracting the second row from the first and using the Taylor expansion $\rho(s) - 1/2 = \sum_{k\geq 0} c_k s^{2k+1}$ with $c_k \neq 0$ (since $\rho(s) - 1/2 = \frac{1}{2}\tanh(s/2)$), comparison of the coefficients of t^{4k+3} yields the system

$$\sum_{h=1}^H z_h(u)^k \Big(A_h(u) + (2k+1) B_h(u) \Big) = 0 \qquad (k \geq 0),$$

where $A_h(u) := \mu_h(u)\, u\, \delta N_h$ and $B_h(u) := \delta\mu_h(u)\, u N_h$. Taking $k = 0, \ldots, 2H - 1$, the resulting scalar coefficient matrix is invertible whenever the $z_h(u)$ are pairwise distinct and nonzero: after column operations it becomes a standard confluent Vandermonde matrix $\begin{bmatrix} z_h(u)^k & k z_h(u)^{k-1} \end{bmatrix}$.

For generic $(M_h, N_h)_{h=1}^H$, the bad set where some μ_h vanishes, some $\mu_h^2 = \mu_j^2$, or some $uN_h = 0$ is a proper algebraic subset of $\mathbb{R}^d$. Hence there exists an open dense set of u for which the linear system forces $A_h(u) = B_h(u) = 0$ for every h, giving $u\,\delta N_h = 0$ and $\delta\mu_h(u) = 0$. Since this holds on an open dense set of u, we conclude $\delta N_h = 0$ and the symmetric part of δM_h is zero.

Step 2: the skew-symmetric part of δM_h also vanishes.

After Step 1, $\delta N_h = 0$ and $\delta M_h^\top = -\delta M_h$. Fix $u, v \in \mathbb{R}^d$ and consider $X_{u,v}(s,t) = (su, tv)$. For skew-symmetric δM_h, $u^\top \delta M_h u = v^\top \delta M_h v = 0$ and $v^\top \delta M_h u = -u^\top \delta M_h v$. A direct differentiation at $t = 0$ gives

$$\partial_t\Big|_{t=0}\Big(\delta\psi_h^{(1)}(X_{u,v}) - \delta\psi_h^{(2)}(X_{u,v})\Big) = -s^2\Big(\rho'(s^2\mu_h(u)) - \frac{1}{4}\Big)\frac{u^\top \delta M_h v}{\sqrt{d_h}}\, uN_h.$$

Summing over h and expanding $\rho'(s) - 1/4 = \sum_{k\geq 1} d_k s^{2k}$ with $d_k \neq 0$, comparing coefficients of s^{4k} yields

$$\sum_{h=1}^{H} z_h(u)^k \frac{u^\top \delta M_h v}{\sqrt{d_h}}\, uN_h = 0 \qquad (k \geq 1).$$

Taking $k = 1, \ldots, H$ and restricting to the open dense set of u for which

$$z_h(u) \neq 0, \qquad z_h(u) \neq z_j(u) \;\; (h \neq j), \qquad uN_h \neq 0 \;\; \text{for all } h,$$

the ordinary Vandermonde matrix on $\{z_h(u)\}_{h=1}^H$ is invertible. Hence each vector coefficient

$$\frac{u^\top \delta M_h v}{\sqrt{d_h}}\, uN_h$$

must vanish. Since $uN_h \neq 0$, this forces

$$u^\top \delta M_h v = 0$$

for every v and every h. Since this holds for an open dense set of u, we get $\delta M_h = 0$.

We have proved $\ker D\widetilde{\Psi}_{(M,N)} = \{0\}$ on a Euclidean-open dense subset of reduced parameter space.

Finally, since $\Psi_{\text{Attn}} = \widetilde{\Psi} \circ \prod_{h=1}^H (q_h \times v_h)$ and the differential of $\widetilde{\Psi}$ is injective on the generic stratum,

$$\ker D\Psi_{\text{Attn}} = \bigoplus_{h=1}^{H} \mathfrak{g}_h^{QK} \;\oplus\; \bigoplus_{h=1}^{H} \mathfrak{g}_h^{VO}, \qquad \dim\ker D\Psi_{\text{Attn}} = H(d_h^2 + d_v^2).$$

Subtracting from the raw parameter count $H(2dd_h + 2dd_v)$ gives $\text{rank}_{\text{gen}}(D\Psi_{\text{Attn}}) = H[d_h(2d - d_h) + d_v(2d - d_v)]$. □

For biases, the same analysis shows:

- query bias contributes $+d_h$ per head;
- key bias contributes 0 because row-wise softmax is invariant under adding a common offset to each row;
- value bias contributes to a constant-output block already represented on the value side;
- output bias contributes a global d-dimensional constant block.

This yields

$$\text{ECC}_{\text{Attn+bias}} = H\big[d_h(2d - d_h + 1) + d_v(2d - d_v)\big] + d.$$

B Proof of Theorem 4.6 (FFN ECC)

Consider the one-hidden-layer ReLU map

$$f(x) = \sum_{i=1}^{m} c_i\, \sigma(a_i^\top x),$$

with $a_i \in \mathbb{R}^d$ and $c_i \in \mathbb{R}^d$. The parameter dimension is $2md$.

Step 1: gauge directions. Because ReLU is positively homogeneous,

$$c_i\,\sigma(a_i^\top x) = (\lambda c_i)\,\sigma((\lambda^{-1}a_i)^\top x) \quad \text{for all } \lambda > 0.$$

Differentiating at $\lambda = 1$ yields one kernel direction per neuron. Hence $\dim\ker D\Psi \geq m$.

Step 2: why pointwise rank is misleading. Fix one activation cone. On that cone the network is affine in x, so the pointwise Jacobian with respect to parameters cannot capture movement of the activation boundaries. Any formula based only on a fixed cone therefore undercounts function-space capacity.

Step 3: distributional jump argument. The derivative of $\sigma(a_i^\top x)$ has a distributional jump across the hyperplane

$$H_i = \{x : a_i^\top x = 0\}.$$

For generic parameters in dimension $d \geq 2$, different neuron hyperplanes are distinct and intersect transversely. A tangent perturbation in the kernel must therefore cancel the singular contributions associated with each H_i separately. This forces each neuron perturbation to preserve both the normal direction of H_i and the output coefficient attached to that boundary. The only remaining infinitesimal symmetry is the positive-scaling gauge from Step 1.

Step 3': explicit jump computation and kernel characterization. Let

$$g_i(x) := \delta c_i\,\sigma(a_i^\top x) + c_i\,1_{\{a_i^\top x>0\}}\,\delta a_i^\top x$$

denote the contribution of neuron i to the first variation. Away from the hyperplane

$$H_i = \{x : a_i^\top x = 0\},$$

the gradient is

$$\nabla_x g_i(x) = \begin{cases} 0, & a_i^\top x < 0, \\ \delta c_i\,a_i^\top + c_i\,\delta a_i^\top, & a_i^\top x > 0. \end{cases}$$

Hence the jump of $\nabla_x g_i$ across H_i is the matrix

$$J_i := \big[\nabla_x g_i\big]_{H_i^+} - \big[\nabla_x g_i\big]_{H_i^-} = \delta c_i\,a_i^\top + c_i\,\delta a_i^\top.$$

Now assume $\delta f = \sum_{i=1}^m g_i \equiv 0$. For generic parameters one has $a_i \neq 0$, $c_i \neq 0$, and the hyperplanes H_i are pairwise distinct. Therefore, for each i, one may choose

$$x_i^\circ \in H_i \setminus \bigcup_{j\neq i} H_j.$$

On a sufficiently small ball B_i around x_i°, every g_j with $j \neq i$ is smooth across $H_i \cap B_i$, so the only jump of $\nabla_x \delta f$ across $H_i \cap B_i$ comes from g_i. Since $\delta f \equiv 0$, its gradient has zero jump. Thus

$$J_i = 0 \qquad (1 \leq i \leq m).$$

It remains to solve

$$\delta c_i\,a_i^\top + c_i\,\delta a_i^\top = 0.$$

Fix i and choose any $w \in a_i^\perp$. Multiplying the above equation on the right by w gives

$$c_i\,(\delta a_i^\top w) = 0.$$

Because $c_i \neq 0$, we obtain $\delta a_i^\top w = 0$ for every $w \in a_i^\perp$. Hence

$$\delta a_i \in \operatorname{span}(a_i),$$

so there exists $\lambda_i \in \mathbb{R}$ such that

$$\delta a_i = -\lambda_i a_i.$$

Substituting back into $J_i = 0$ yields

$$(\delta c_i - \lambda_i c_i) a_i^\top = 0.$$

Since $a_i \neq 0$, we conclude

$$\delta c_i = \lambda_i c_i.$$

Therefore every kernel vector is of the form

$$(\delta c_i, \delta a_i) = \lambda_i (c_i, -a_i), \qquad i = 1, \ldots, m,$$

which is exactly the infinitesimal positive-homogeneity gauge from Step 1. The singular parts separate neuron-by-neuron, and

$$\ker D\Psi = \bigoplus_{i=1}^{m} \mathbb{R}\,(c_i, -a_i), \qquad \dim \ker D\Psi = m.$$

This supplies the missing kernel calculation and completes the proof of

$$\mathrm{ECC}_{\mathrm{FFN}} = 2md - m = m(2d-1).$$

Therefore the generic kernel is exactly m-dimensional, so

$$\mathrm{ECC}_{\mathrm{FFN}} = 2md - m = m(2d-1).$$

Biases. Including hidden and output biases raises the raw parameter count to

$$md + m + dm + d = 2md + m + d.$$

The same m gauge directions remain, so

$$\mathrm{ECC}_{\mathrm{FFN+bias}} = (2md + m + d) - m = 2md + d.$$

C Proof of Theorem 4.16 (Mamba ECC)

Standing assumptions for Appendix C. Throughout this appendix, "standard Mamba-v1 block" means the following concrete realization on input sequences $u_{1:L} \in (\mathbb{R}^d)^L$:

(C1) The input is split into an x-branch and a gating z-branch,

$$x_t^{\mathrm{pre}} = W_x^\top u_t, \qquad z_t = W_z^\top u_t,$$

with $W_x, W_z \in \mathbb{R}^{d\times m}$, and the block ends with the standard output map

$$o_t = W_o^\top \big(\mathrm{silu}(z_t) \odot y_t\big), \qquad W_o \in \mathbb{R}^{m\times d},$$

where

$$\mathrm{silu}(s) := \frac{s}{1+e^{-s}}.$$

(C2) The x-branch is processed by a depthwise causal convolution of width k, followed by the standard pointwise SiLU nonlinearity:

$$\widetilde{x}_{t,i} := \sum_{\ell=0}^{k-1} K_{\ell,i}\, x_{t-\ell,i}^{\mathrm{pre}} + b_{\mathrm{conv},i}, \qquad x_{t,i} := \mathrm{silu}(\widetilde{x}_{t,i}),$$

with $x_s^{\mathrm{pre}} = 0$ for $s \leq 0$. In the remainder of Appendix C, x_t denotes this post-convolution/post-SiLU signal.

(C3) The step-size channel has rank r: there exist $U, V \in \mathbb{R}^{m\times r}$ such that

$$T = UV^\top \in \mathbb{R}^{m\times m},$$

and $b_\Delta \in \mathbb{R}^m$ is its bias.

(C4) The per-channel step size is parameterized by softplus,

$$\Delta_{t,i} = \mathrm{softplus}\big((Tx_t)_i + (b_\Delta)_i\big), \qquad \mathrm{softplus}(s) := \log(1 + e^s).$$

(C5) The state matrix is diagonal in the state coordinate, so the selective-scan exponents are $A_{i,n}$ with

$$A = (A_{i,n})_{1\le i\le m,\ 1\le n\le N} \in \mathbb{R}^{m\times N}.$$

(C6) The state input/output maps factor through rank-one kernels

$$S_n = b_n c_n^\top, \qquad b_n, c_n \in \mathbb{R}^m, \qquad 1 \le n \le N.$$

(C7) The selective-scan core is

$$h_{t,i,n} = e^{A_{i,n}\Delta_{t,i}}\, h_{t-1,i,n} + \Delta_{t,i}(x_t^\top b_n)\, x_{t,i},$$

$$y_{t,i} = D_i x_{t,i} + \sum_{n=1}^{N} (x_t^\top c_n)\, h_{t,i,n},$$

with skip vector $D \in \mathbb{R}^m$.

(C8) The initial state is zero,

$$h_{0,i,n} = 0,$$

and the sequence length satisfies

$$L \ge \max(2, k).$$

(C9) The partial differential with respect to the front/output parameters

$$(W_x, W_z, W_o, K_0, \dots, K_{k-1}, b_{\mathrm{conv}})$$

is injective, and its image is transverse to (equivalently, intersects trivially with) the image of the partial differential with respect to the reduced selective-scan parameters

$$(T, b_\Delta, A, D, S_1, \dots, S_N).$$

(C10) For each inner channel $i \in \{1, \dots, m\}$ there exists a nonempty open set $V_i \subset \mathbb{R}^m$ such that, for every $v \in V_i$, the $4N$ functions

$$e^{A_{i,n}\Delta_i(s;v)}, \quad \Delta_i(s;v)e^{A_{i,n}\Delta_i(s;v)}, \quad \eta_i(s;v)e^{A_{i,n}\Delta_i(s;v)}, \quad s\,\eta_i(s;v)e^{A_{i,n}\Delta_i(s;v)}$$

for $n = 1, \dots, N$ are linearly independent on some interval, where

$$\Delta_i(s;v) := \mathrm{softplus}\big((Tv)_i\, s + (b_\Delta)_i\big), \qquad \eta_i(s;v) := \mathrm{softplus}'\big((Tv)_i\, s + (b_\Delta)_i\big).$$

Assumptions (C9)–(C10) are the only genuinely genericity/transversality inputs; Proposition C.1 below shows that they hold on a Euclidean-open dense subset of the structural parameter space determined by (C1)–(C8).

Proposition C.1 (Genericity of Assumptions (C9)–(C10))**.** *Fix the structural class determined by Assumptions (C1)–(C8), and let Θ_{str} be its parameter space. Then the subset of Θ_{str} on which Assumptions (C9)–(C10) hold is Euclidean-open dense.*

Proof. We treat (C10) and (C9) in turn.

Genericity of (C10). Fix a channel i. If the i-th row T_i is nonzero, then

$$V_i := \{v \in \mathbb{R}^m : (Tv)_i \neq 0\}$$

is nonempty and open. For $v \in V_i$, write

$$a_i(v) := (Tv)_i \neq 0, \qquad b_i := (b_\Delta)_i,$$

and set

$$t := e^{a_i(v)s+b_i} > 0.$$

Then

$$\Delta_i(s; v) = \log(1+t), \qquad \eta_i(s; v) = \frac{t}{1+t},$$

so the four families in (C10) become

$$f_{1,n}(t) := (1+t)^{A_{i,n}}, \qquad f_{2,n}(t) := \log(1+t)\,(1+t)^{A_{i,n}},$$
$$f_{3,n}(t) := t(1+t)^{A_{i,n}-1}, \qquad f_{4,n}(t) := (\log t)\,t(1+t)^{A_{i,n}-1}.$$

Since $a_i(v) \neq 0$, one can choose a tail interval in s on which t is arbitrarily large. As $t \to \infty$,

$$\begin{aligned}
f_{1,n}(t) &= t^{A_{i,n}} + A_{i,n}t^{A_{i,n}-1} + O(t^{A_{i,n}-2}),\\
f_{2,n}(t) &= (\log t)t^{A_{i,n}} + A_{i,n}(\log t)t^{A_{i,n}-1} + t^{A_{i,n}-1} + O((\log t)t^{A_{i,n}-2}),\\
f_{3,n}(t) &= t^{A_{i,n}} + (A_{i,n}-1)t^{A_{i,n}-1} + O(t^{A_{i,n}-2}),\\
f_{4,n}(t) &= (\log t)t^{A_{i,n}} + (A_{i,n}-1)(\log t)t^{A_{i,n}-1} + O((\log t)t^{A_{i,n}-2}).
\end{aligned}$$

Assume that

$$A_{i,n} \neq A_{i,m}, \qquad A_{i,n} - A_{i,m} \notin \{-1, 1\} \qquad (n \neq m).$$

Suppose

$$\sum_{n=1}^{N} \big(\alpha_n f_{1,n} + \beta_n f_{2,n} + \gamma_n f_{3,n} + \delta_n f_{4,n}\big) \equiv 0.$$

Pick $n_\star$ so that $A_{i,n_\star} = \max_n A_{i,n}$. Because no other exponent equals $A_{i,n_\star}$ or $A_{i,n_\star} - 1$, comparison of the coefficients of $(\log t)t^{A_{i,n_\star}}$ and $t^{A_{i,n_\star}}$ gives

$$\beta_{n_\star} + \delta_{n_\star} = 0, \qquad \alpha_{n_\star} + \gamma_{n_\star} = 0.$$

Comparing next the coefficients of $(\log t)t^{A_{i,n_\star}-1}$ and $t^{A_{i,n_\star}-1}$ yields

$$A_{i,n_\star}\beta_{n_\star} + (A_{i,n_\star}-1)\delta_{n_\star} = 0, \qquad A_{i,n_\star}\alpha_{n_\star} + (A_{i,n_\star}-1)\gamma_{n_\star} = 0,$$

and therefore

$$\alpha_{n_\star} = \beta_{n_\star} = \gamma_{n_\star} = \delta_{n_\star} = 0.$$

Removing this index and iterating downward in the ordered list of exponents shows that all coefficients vanish. Hence the $4N$ functions are linearly independent on every tail interval on which t is sufficiently large, so (C10) holds.

The failure conditions are precisely that some row T_i vanish or that one of the finitely many equalities

$$A_{i,n} = A_{i,m}, \qquad A_{i,n} = A_{i,m} \pm 1$$

hold. Their union is a proper algebraic subset. Thus (C10) holds on a Euclidean-open dense subset of Θ_{str}.

Genericity of (C9). Fix a finite probe family $\mathcal{X}$ consisting of: (i) one-token coordinate impulses, (ii) two-token coordinate impulses, (iii) lag probes up to time k, and (iv) one constant nonzero baseline sequence. For each

parameter point θ, collect the first and second input-amplitude jets of the corresponding block outputs on $\mathcal{X}$ into a finite matrix

$$\mathfrak{J}(\theta) = \big[\mathfrak{J}_{\mathrm{fo}}(\theta) \;\; \mathfrak{J}_{\mathrm{ss}}(\theta)\big],$$

whose columns are indexed respectively by the front/output parameters

$$(W_x, W_z, W_o, K_0, \ldots, K_{k-1}, b_{\mathrm{conv}})$$

and the reduced selective-scan parameters

$$(T, b_\Delta, A, D, S_1, \ldots, S_N).$$

Because the block formulas in Assumptions (C1)–(C8) are real-analytic, every entry of $\mathfrak{J}(\theta)$ is real-analytic in θ.

Choose $\mathcal{X}$ so that, on the constant baseline probe, the vectors

$$\mathrm{silu}(z_t) \odot y_t$$

span $\mathbb{R}^m$, and on the impulse/lag probes the pre-scan jets

$$(\widetilde{x}_t, z_t)_{t=1}^{k+1}$$

separate the parameters of

$$(W_x, W_z, K_0, \ldots, K_{k-1}, b_{\mathrm{conv}})$$

by channel and by lag. Order the rows and columns of $\mathfrak{J}(\theta)$ lexicographically by signature (pre-scan time, lag, gate/readout, input degree). At any point where the corresponding probe minors are nonzero, the front/output block $\mathfrak{J}_{\mathrm{fo}}(\theta)$ is block triangular with nonzero diagonal blocks:

- the lag-ℓ block isolates K_ℓ $(1 \le \ell \le k-1)$;
- the time-1 pre-scan block isolates W_x, K_0, and b_{conv};
- the gate block on the baseline probe isolates W_z because $(\mathrm{silu})'$ is nonzero on an open set;
- the readout block on the baseline probe isolates W_o because the vectors $\mathrm{silu}(z_t) \odot y_t$ span $\mathbb{R}^m$.

Hence the partial differential with respect to the front/output parameters is injective.

With the same ordering, reduced selective-scan directions are identically zero on the pre-scan jet blocks, whereas every nonzero variation of

$$(W_x, W_z, K_0, \ldots, K_{k-1}, b_{\mathrm{conv}})$$

changes at least one entry of those blocks. The W_o-columns occupy an additional readout block on the baseline probe. Consequently a mixed minor of $\mathfrak{J}(\theta)$ is block triangular with nonzero diagonal blocks, so

$$\operatorname{rank} \mathfrak{J}(\theta) = \operatorname{rank} \mathfrak{J}_{\mathrm{fo}}(\theta) + \operatorname{rank} \mathfrak{J}_{\mathrm{ss}}(\theta).$$

This is exactly the direct-sum/transversality statement in (C9).

The relevant minors are real-analytic and nonzero at one generic point (take generic full-rank W_x, W_z, W_o, nonzero convolution taps, nonzero D, and a baseline probe with all channel responses nonzero), so their common nonvanishing locus is Euclidean-open dense. Therefore (C9) also holds on a Euclidean-open dense subset of Θ_{str}. □

Lemma C.2 (Kernel exactness for the standard Mamba block)**.** *Assume Standing Assumptions (C1)–(C10) above. Then the Jacobian kernel of the full realization map is exactly the direct sum of the $\mathfrak{gl}(r)$-gauge of the step-size factorization and the N independent state-coordinate rescalings*

$$b_n \mapsto \lambda_n b_n, \qquad c_n \mapsto \lambda_n^{-1} c_n \qquad (1 \le n \le N).$$

In particular,

$$\dim\ker D\Psi_{\text{Mamba}} = r^2 + N$$

on a Euclidean-open dense subset of parameter space, and therefore

$$\operatorname{rank}_{\text{gen}}(D\Psi_{\text{Mamba}}) = 3dm + m(k + 3N + 2r + 3) - r^2 - N.$$

Proof. The full parameterization factors through the reduced coordinates $T, b_\Delta, A, D, S_1, \ldots, S_N$. The two quotient maps are:

- the rank-r factorization, whose fibers are the $\mathrm{GL}(r)$-orbits; tangent kernel dimension r^2;
- the coordinatewise rescaling $(b_n, c_n) \mapsto (\lambda_n b_n, \lambda_n^{-1} c_n)$; tangent kernel dimension N.

Thus it suffices to prove that the differential of the reduced realization map is injective.

By Standing Assumption (C9), it is enough to prove injectivity for the reduced selective-scan core. Write $\Delta_{t,i} := \mathrm{softplus}((Tx_t)_i + (b_\Delta)_i)$ and let the core recurrence be

$$h_{t,i,n} = e^{A_{i,n}\Delta_{t,i}}\, h_{t-1,i,n} + \Delta_{t,i}\,(x_t^\top b_n)\, x_{t,i}, \qquad y_{t,i} = D_i x_{t,i} + \sum_{n=1}^{N} (x_t^\top c_n)\, h_{t,i,n}.$$

Step 1: $\delta D = 0$**.** Take the one-token slice $x_1 = \varepsilon u$, $x_t = 0$ for $t > 1$. Then $h_{1,i,n} = O(\varepsilon^2)$ and $y_{1,i} = D_i \varepsilon u_i + O(\varepsilon^3)$. The coefficient of ε in $\delta y_{1,i}$ must vanish, hence $\delta D_i = 0$.

Step 2: $\delta T = \delta b_\Delta = \delta A = \delta S_n = 0$**.** Fix a channel i, choose $v \in V_i$, and consider the two-token slice $x_1 = \varepsilon u$, $x_2 = sv$, $x_t = 0$ for $t > 2$. Expanding the lag-one term at order $\varepsilon^2 s$, the coefficient of $\varepsilon^2 s u_i$ in $y_{2,i}$ is

$$\Gamma_{i,u,v}(s) = \bar{\Delta}_i \sum_{n=1}^{N} \omega_n(u,v)\, e^{A_{i,n}\Delta_i(s;v)},$$

where $\bar{\Delta}_i := \mathrm{softplus}((b_\Delta)_i)$ and $\omega_n(u,v) := u^\top S_n v$. Since $\delta\Gamma_{i,u,v}(s) \equiv 0$, differentiating and using $\delta\Delta_i(s;v) = \eta_i(s;v)\,\delta(b_\Delta)_i + s\,\eta_i(s;v)\,(\delta T_i v)$, the first variation becomes a linear combination of the four function families in Standing Assumption (C10). Linear independence forces every coefficient to vanish. For generic (u,v) with $\omega_n(u,v) \neq 0$, this gives

$$\delta\bar{\Delta}_i = 0, \quad \delta A_{i,n} = 0, \quad \delta(b_\Delta)_i = 0, \quad \delta T_i v = 0, \quad u^\top \delta S_n v = 0.$$

Varying u over an open dense set and v over V_i gives $\delta S_n = 0$ and $\delta T_i = 0$ for every i.

Step 3: completion. By Step 2 the reduced selective-scan core differential is injective. Standing Assumption (C9) then ensures the full reduced differential is injective. Lifting through the quotient maps gives

$$\dim\ker D\Psi_{\text{Mamba}} = r^2 + N.$$

□

Substituting the default hyperparameters $m = 2d$, $N = 16$, $k = 4$, and $r = \lceil d/16 \rceil$ yields

$$\begin{aligned}
\mathrm{ECC}_{\text{Mamba}} &= 3d(2d) + 2d(4 + 3\cdot 16 + 2r + 3) - r^2 - 16 \\
&= 6d^2 + 110d + 4dr - r^2 - 16 \\
&= 6d^2 + \frac{63}{256}d^2 + O(d) \\
&= \frac{1599}{256}d^2 + O(d) \approx 6.246\, d^2 + O(d).
\end{aligned}$$

D Coverage, efficiency, and the no-universal-optimum theorem

This appendix records the formal distinction between coverage and efficiency.

Let A be an architecture with capacity $C_A = \mathrm{ECC}(A)$ and let D be a task family with effective dimension k_D. Define

$$k_{\mathrm{use}}(A, D) = \sup_{\theta} \operatorname{erank}(M_{A,D}(\theta)).$$

Then

$$\gamma_{\mathrm{cov}}(A, D) = \frac{k_{\mathrm{use}}(A, D)}{k_D}, \qquad \eta_{\mathrm{eff}}(A, D) = \frac{k_{\mathrm{use}}(A, D)}{C_A}.$$

Coverage asks whether the relevant task directions are reachable at all. Efficiency asks how much architectural capacity is wasted on irrelevant directions. These quantities can move in opposite directions.

Theorem 8.3 is then immediate: if two architectures exchange their efficiency ordering across two domains, the maximizing architecture for expected efficiency depends on the domain mixture. This formalizes the claim that universal coverage and specialist efficiency are distinct design goals.

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